An Archival ALMA Search for Unresolved Spectral-Carrier Technosignatures toward Stars within 40 pc at 114–873 GHz

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ABSTRACT

We search 405 fine-channel ALMA windows toward 61 systems within 40 pc, covering 47.7 GHz of union bandwidth in 26 disjoint intervals, for narrowband drifting carriers; 1258 coarser windows constrain only an unresolved excess. Injection gives a median single-epoch sensitivity, at ninety per cent recovery, of 1.8×10^{15} W equivalent isotropic power for a carrier narrower than the instrumental channel. We find 56 threshold crossings (channels and drift rates where the star reaches 5 times the scatter at control positions), 4 in one block failing the quality criterion. Of the other 52, 18 coincide with catalogued molecular transitions, 16 with corroborating evidence of line emission. The 31 unattributed crossings of the primary census are consistent with chance against the 24.8 expected at control positions ($P = 0.14$), which does not exclude a smaller added population; a narrower field gives 35 against 24.1 ($P = 0.027$ before correcting for that choice). No confirmed carrier is found. Coverage and the criterion, not depth, limit confirmation: 21 systems have no repeat at a covered frequency, and only a carrier returning to its discovery frequency could be confirmed.

Subject headings: Technosignatures, Search for extraterrestrial intelligence, Submillimeter astronomy, Radio spectroscopy, Debris disks

1. INTRODUCTION

The public ALMA archive holds thousands of hours of millimetre and submillimetre observations taken for other reasons, mostly studies of dust discs and planet formation. This paper searches them for a narrowband transmitter at the position of a star in the field. No such signal was found.

Searches for *technosignatures* – detectable imprints of technology, intended or not (Sheikh 2020; Vidal et al. 2026) – have examined many thousands of stars over six decades, from the first proposals (Cocconi & Morrison 1959; Drake 1961) through Project Phoenix (Tarter 2001; Backus & Project Phoenix Team 2002) to the targeted and commensal surveys of today (Zhang et al. 2020; Czech et al. 2021; Li et al. 2026; Johnson et al. 2023; Choza et al. 2024; Morrison et al. 2023; Tremblay & Tingay 2024; Manunza et al. 2025), now centred on machine-learning interference rejection (Ma et al. 2023; Jacobson-Bell et al. 2025; Pardo et al. 2025). The last published accounting of that effort covers only 6×10^{-18} of the space of transmitter position, power, frequency, bandwidth, polarisation and repetition rate (Wright et al. 2018). Essentially all of it lies below ∼30 GHz.

Above about 30 GHz the record is nearly empty, and that is what makes this frequency range new. Steffes & DeBoer (1994) observed 40 solar-type stars within 23 pc at 203 GHz and Mauersberger et al. (1996) included τ Cet and ϵ Eri among 17 targets at the same frequency, each covering no more than 35 MHz of a single line – 1363 times less than the union searched here – with no search over drift rate and no epoch-to-epoch test of persistence. Of ALMA's receiver bands only Band 3 has been searched for technosignatures (Mason et al. 2025).[4]

At these frequencies interstellar dispersion and scintillation are negligible and an aperture delivers far more gain per watt, but system temperatures are higher, the atmosphere is opaque in places at both ends of the link and the narrow beam – 25 arcsec for the 12 m benchmark of §5.1 – must be aimed at the Earth deliberately, so limits are stated as EIRP, which needs no transmitter model. The archive adds limitations of its own: the depth and resolution of each spectrum are the original observer's; only some systems have a second, separated epoch, so what could be searched and what could be confirmed differ; and the targets were chosen by other programmes, so the sample represents no population and supports no bound on the fraction of stars hosting a transmitter (Wright et al. 2018; Margot et al. 2023; Sheikh et al. 2025) (§2.1).

The signal sought is an *unresolved spectral carrier*: emission confined to a few channels, unresolved at the stellar position, allowed to drift as it must for a transmitter on a rotating, orbiting body (§3.1), and present at more than one epoch. "Narrowband" describes the assumed transmitter and not the resolving power of the archived channels, and broadband planetary-scale leakage (Garrett 2026) is outside the search by construction. Recurrence, and not brightness, separates an interesting event from a detection, and a signal of interest needs detailed follow-up (Sheikh et al. 2021).

The method meets the obstacles Mason et al. (2025) identified – drift rates that scale with frequency, smearing unless drift is searched for, and the molecular line forest – with a drift search, a line catalogue against which every event is compared, and a recurrence test in a separated epoch; and in place of interference rejection it tests

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[4] The 35–50 GHz band holds observations of 4 stars of this sample, all proprietary until 2027 and all in a continuum mode whose channels are 2048 times coarser than the narrowest channel searched here, so they cannot carry a carrier search; the band above it holds none.

the star against control positions inside its own primary beam, which a single dish cannot resolve. Every window was re-reduced and re-searched from the raw archive (§3).

Experiments are compared here on the minimum detectable received flux $F_{\rm min} = S_{\rm min}\,\delta\nu$ at each one's own channel width: $F_{\rm min}$ rises as $\delta\nu^{1/2}$, so a coarse window put on a fine channel would be credited with a depth it never reached. A limit also holds only for a signal narrower than the native channel that set it, and those channels span 7 orders of magnitude, from 2.79 Hz to 31.2 MHz. Figure 1 draws each published threshold beside its value rescaled to a common 5σ (3 of the 4 programmes need it), an arithmetic convention and not a matched false-alarm probability (Appendix D); the values, with each programme's distance, are in `figures/figure_data.csv` of the data archive (White & Dey 2026). All are nominal thresholds except this survey's EIRP_{90} (open symbols), measured with injected carriers – the step Price et al. (2020) name as missing from their own limits and Margot et al. (2023) measure but do not fold into theirs. None carries a confirmation criterion, and the 24 stars drawn as grey triangles, covered in coarse channels alone, constrain an unresolved excess and not a drifting carrier. Mason et al. reach a median $4.9 \times 10^{-23}\,\mathrm{W\,m^{-2}}$ (3.8×10^{-23}–5.8×10^{-23} across their four fields) in a 30.52 kHz channel, against $1.2 \times 10^{-23}\,\mathrm{W\,m^{-2}}$ in a 122.1 kHz channel for the deepest window that lies outside the line mask – the only bandwidth this survey constrains (§3.2) – both at a bare 5σ threshold, a factor 4 deeper than their median field; the median window here is $3.2 \times 10^{-22}\,\mathrm{W\,m^{-2}}$ in a 488 kHz channel, a factor 7 shallower. Per channel this survey is the shallower of the two except in its finest windows; what distinguishes it (panel b) is its frequency range and its 61 systems within 40 pc, against distant stars in 4 calibrator fields.

2. THE SAMPLE

This search covers 1663 spectral windows toward 90 stars in 83 systems within 40 pc. They are drawn from 407 public ALMA execution blocks and 4.1 TB of raw data put through one pipeline: 281 h of telescope time, and 1098 h summed over windows. That set is the *primary census*, over which every sensitivity, every expectation and every limit here is computed. It holds 50 of the 56 threshold crossings this paper reports; 4 more are coarse-channel crossings of the same census in the one execution block that fails the data-quality criterion of §3.2; and the remaining 2 lie in an *archival tail* of blocks searched after the census was closed, toward stars it already holds, which are not tabulated window by window and enter no expectation. Three further sets – an *archival extension* of blocks toward the same stars, a reserved *hold-out* set aside by the identifier rule of Appendix B.1, and a control beyond 40 pc – enter no census statistic (§4.6.1). Table A1 names every set in rows that sum, and Table A2 resolves every count of windows, crossings and systems into the set it is counted over. *Star* means one stellar component, *system* a bound group counted once however many of its components were observed, and *target* one ALMA pointing.

The census is two experiments, divided on native channel width, which is what the search statistic and the completeness calibration both depend on. *Class A* is the 405

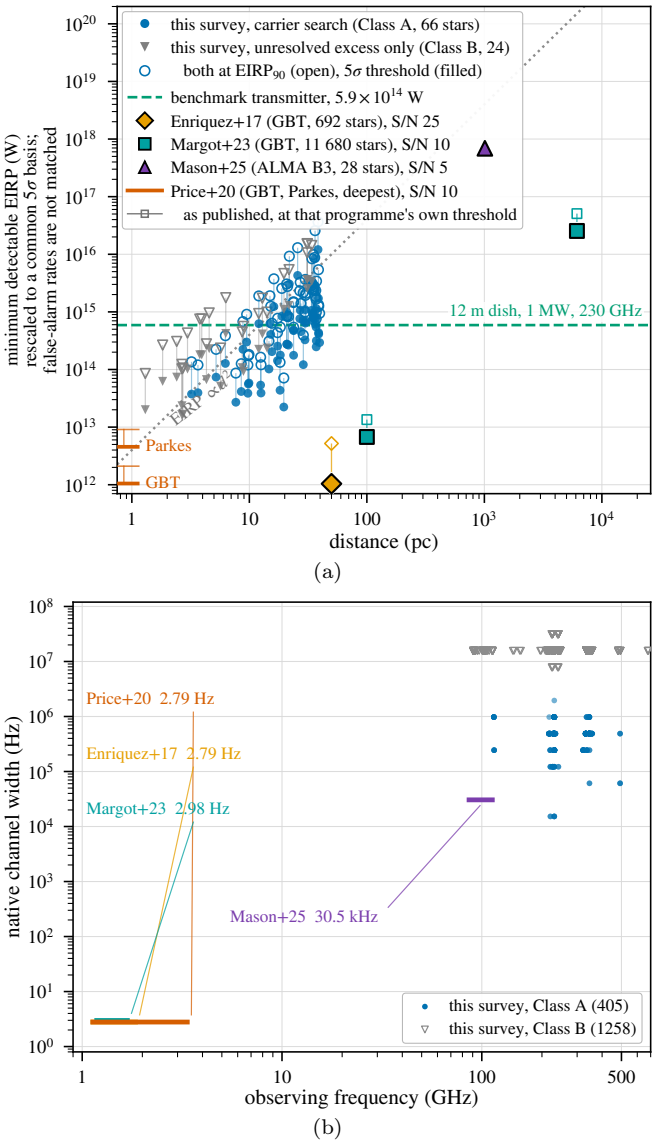


FIG. 1.— Where this survey sits among published radio technosignature searches (§1). **(a)** Minimum detectable EIRP against distance; this survey's deepest window per star – blue, the carrier search; grey, coarse channels only; filled, the bare threshold; open, EIRP_{90}. **(b)** Native channel width against sky frequency (filled, Class A; open, Class B).

fine-channel windows toward 66 stars in 61 systems, of which 391 are channelised finely enough for the searched drift to cross a channel and 14 are not (§3.1); every sensitivity and limit quoted here refers to Class A unless stated otherwise. *Class B* is the remaining 1258 coarse-channel windows, which support a secondary search for an unresolved spectral excess. The channelisation is bimodal, so the division is unambiguous: the widest fine channel is 1.95 MHz and the narrowest coarse one 7.81 MHz, a factor of 4 with nothing between them.

Every inclusion criterion was applied by query. A star enters if its Gaia DR3 parallax is at least 25 mas, measured to better than 20 per cent (a clause that never binds, the worst fractional error among the stars searched being 2.5 per cent), and if its proper-motion-corrected position at the observation epoch falls inside the quoted field of view of at least one public execution block. Dis-

tances are direct inversions, $d = 1/\varpi$, and the star need not sit at the pointing centre. A window is kept only where the primary-beam response at the star is at least 0.22, the offset at which the adopted Gaussian and a blocked-aperture response part by 10 per cent; the rule removes no window, the furthest lying at $0.82\,\lambda/D$ against a limit at $0.83\,\lambda/D$ (Appendix A.2). The quoted field of view is optimistic: of the 168 work-list entries the query returns, only 115 stars prove to be contained by a public field, and 90 of those yielded a searchable window. One target, a serendipitous field star, appears throughout under the designation of the ALMA source at its field centre, ALMA J153702653−33192492; it is 2MASS J15370316−3319273 (Gaia DR3 6015967739554000768), an M5 dwarf at 35.0 pc with no radial velocity in SIMBAD, VizieR or *Gaia* DR3.

Disc, planet and colour enter as descriptive metadata only. A star with no Gaia DR3 parallax cannot enter, and every one of the 22 stars in this volume brighter than second magnitude lacks one, so 257 of the 4227 Hipparcos stars in this volume whose own parallaxes meet the criteria never reached the work list – not only the brightest, 198 of them being fainter than Hp 4 – and seven of them, Sirius A, α Centauri A, Vega, Fomalhaut, α Centauri B, β Leonis and MCC 124, lie inside a public field holding 38 fine-channel and 374 coarse-channel windows, 1.0 TB of raw data in 83 execution blocks, none of it searched here. A search of this volume should admit these stars on Hipparcos astrometry where Gaia has none.

2.1. *The archival selection*

This sample was not designed: it is whatever previous ALMA programmes happened to observe among the stars Gaia DR3 catalogues, and 90 per cent of the stars searched were targeted under a debris-disc or planet-formation proposal. Table 1 measures the bias against a volume-limited reference census of which the searched stars are a subset, both columns the same Gaia DR3 selection – $\varpi > 0$, $\sigma_\varpi/\varpi \leq 0.1$ and $1/\varpi \leq 40$ pc, 17 566 objects – and both classified by one rule, BP−RP colour setting the class and absolute G magnitude separating the degenerates. It reaches 17 278 census objects and 88 of the 90 searched stars, the other 2 carrying no Gaia BP or RP photometry; neither column contains the stars Gaia lacks, so the ratios are internally consistent without them.

Distance is the dominant selection effect: stars within 5 pc are over-represented by a factor 53, and the enhancement falls monotonically with distance to a factor of two *under*-representation beyond 30 pc. Colour comes second, A stars by about 17 and F and G stars by factors of three, because those are the stars with bright debris discs. For the A stars that factor is a floor. The Gaia requirement is itself colour-selective, removing stars as blue as an A star at 3.7 times its rate among all the others. M dwarfs are depleted: 41 of the 90 searched stars, a factor 0.7 of their share of the reference census, and only 28 of them carry fine-channel coverage at all. Coverage is concentrated too, 285 of the 1663 windows coming from 3 systems. No statement in this paper is therefore an occurrence rate for the solar neighbourhood. The selection acts hardest on exactly the targets a designed search would pick. Of the 16 sample systems within 15 pc with a confirmed planet, 6 carry no Class A coverage and so no drift-resolving limit.

TABLE 1
Selection function of the 90 searched stars against the volume-limited Gaia DR3 census of §2.1. "Ratio": the searched over the census column fraction.

Property	Census	Searched	Ratio
Distance			
$d \leq 5$ pc	48 (0.3%)	13 (14.4%)	52.9
5–10 pc	240 (1.4%)	13 (14.4%)	10.6
10–20 pc	2069 (11.8%)	18 (20.0%)	1.7
20–30 pc	5391 (30.7%)	19 (21.1%)	0.7
30–40 pc	9818 (55.9%)	27 (30.0%)	0.5
Colour class[a]			
A	61 (0.4%)	5 (6.2%)	17.2
F	473 (2.8%)	8 (10.0%)	3.6
G	978 (5.8%)	15 (18.8%)	3.2
K	2221 (13.2%)	11 (13.8%)	1.0
M	13 082 (77.8%)	41 (51.2%)	0.7
Other axes			
Confirmed planet hosts[b]	26 (15.5%)	24 (26.7%)	1.7
White dwarfs[a]	463 (2.7%)	8 (9.1%)	3.4
Searched stars / systems	–	90 / 83	–

[a] Colour bins, not MK types. The boundaries are the colours at which the census's own temperature–colour relation crosses 7500, 6000, 5300 and 3900 K: BP−RP < 0.36 for A, 0.36–0.72 for F, 0.72–0.95 for G, 0.95–1.77 for K and > 1.77 for M. Over the 8594 census objects that also carry a Gaia temperature the two classifications agree for 98 per cent, and every disagreement is between neighbouring bins. A star is counted as degenerate when it lies more than 4 mag below the main sequence measured from the census itself: the two sequences are separated by 5 to 8 mag here, so moving the cut by a magnitude either way changes the count by under 3 per cent. The five classes are normalised over the 80 non-degenerate searched stars and the 16 815 non-degenerate census objects, and the degenerate row over the 88 classified searched stars and the 17 278 classified census objects, which is the same rule in both columns.

[b] Both columns are the same positional join, of the 168-entry ALMA-covered work list and of its searched subset, to the NASA Exoplanet Archive `pscomppars` table accessed 2026 October 4. The 24 searched hosts carry 51 confirmed planets, 22 of the 24 surviving the deuterium-burning mass limit; the Gaia census carries no planet information, so the census column in this row is the work list and not the 17 566-object Gaia census.

2.2. *Frequency coverage and archival scope*

The Class A experiment covers $\Delta\nu_A = 47.7$ GHz of unique sky frequency, and that coverage is sparse: 26 disjoint islands between 114.0 and 873.1 GHz instead of a band, so those endpoints bound where observations exist and not what was covered. With the coarse-channel windows the archival coverage is 118.1 GHz in 46 intervals, of which only 114.4 GHz survives the molecular-line mask of §3. Fig. 2 shows the distribution: a transmitter between the islands was never observable here, whatever its power.

The archive is not exhausted either: 177 further blocks lie inside the same query and were unprocessed when the census was frozen. Table A1 accounts for every one of them row by row, including the 115 reported here as the archival extension. The largest single gap is the 18 blocks whose raw visibilities the archive holds but never pipeline-calibrated, 237 GB in all. No effort of ours would close that gap.

3. DATA AND METHODOLOGY

The search looks for one kind of signal: an unresolved carrier at the star's own position, narrow compared with a single correlator channel, and drifting in sky

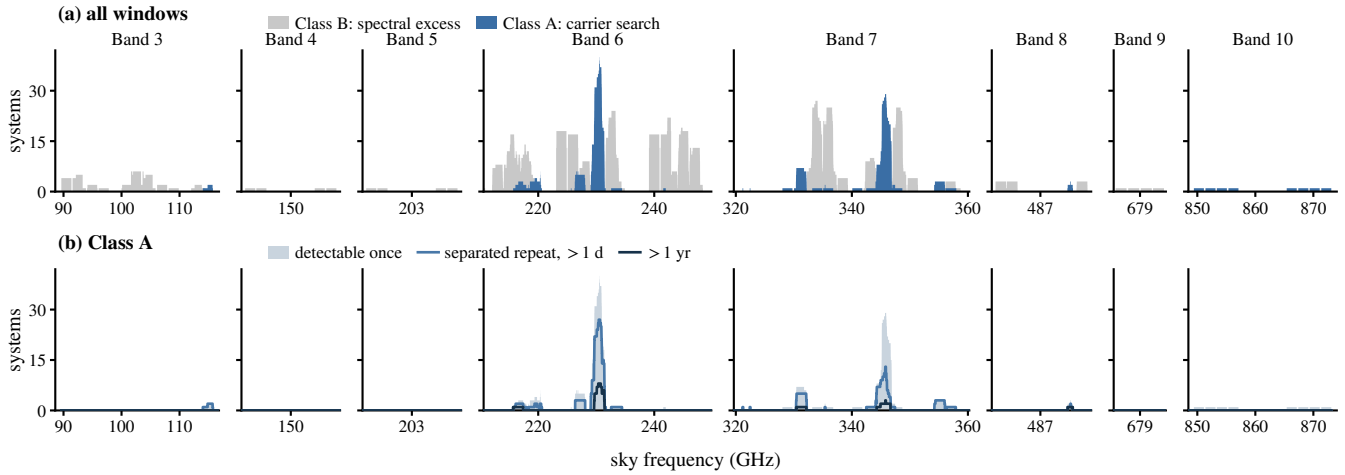


FIG. 2.— **Where this survey searched, and where a repeat could test it**: independent systems covering each sky frequency, one panel per populated ALMA band, its width proportional to the band's frequency share; curves step at the searched window edges themselves. *(a)* Every window: blue, the fine-channel carrier search (Class A); grey, the coarse-channel spectral-excess search (Class B); the summed window bandwidth is 24.1× the union. *(b)* Class A: detectable once (grey); with a *separated* block (Table 2) covering the frequency (blue); beyond a year (dark). An accelerating carrier further needs a matching orbital phase.

frequency because the transmitter accelerates along the line of sight. Over a long track its power is spread across many channels, so a search that treats each channel on its own will miss it. The analysis therefore *de-drifts* the data before it measures them. It shifts each integration's spectrum to follow a trial drift rate and only then adds them together.

3.1. *The de-drifted search statistic*

Each window's dynamic spectrum is stacked along every trial drift rate $\dot{\nu}$. The search statistic is the largest stacked signal-to-noise ratio anywhere on the resulting channel×drift grid,

$$T(\mathbf{x}) = \max_{\dot{\nu},\,\nu} \frac{\sum_t I(\mathbf{x}, t, \nu + \dot{\nu}t) \,/\sigma^2(t, \nu + \dot{\nu}t)}{\left[\sum_t 1/\sigma^2(t, \nu + \dot{\nu}t)\right]^{1/2}}, \qquad (1)$$

where $I(\mathbf{x}, t, \nu)$ is the spectrum at sky position $\mathbf{x}$ in integration t and channel ν, and $\sigma(t, \nu)$ is the noise scale there. Both weights are evaluated at the channel the track occupies in that integration, $\nu + \dot{\nu}t$, and not at the channel that labels it. Equation 1 is the inverse-variance-weighted stack of the calibrated intensities along a constant-drift track, divided by the nominal noise of that stack. It is the matched filter for such a carrier, and T an absolute signal-to-noise ratio, only when the noise is independent between integrations and between channels, an idealisation for which no optimality is claimed. The trigger is therefore set against the distribution T takes in the data, given empirically in every window by 512 control positions on the same visibilities and through the same stack, so any correlation the noise carries is inside them; and a value of T is converted into a power by injected carriers recovered through the unchanged pipeline (§4.1), without passing through the independence assumption at all. In Equation 1 the time t runs from the first integration of the observation, so the frequency ν that labels a track is the one a carrier would occupy at the start of that block, and a repeat epoch is read at the cell the same convention fixes (§4.6). The noise scale is the median absolute deviation across 512 control positions at each integration and each channel, scaled to its Gaussian-equivalent standard deviation: local in time and frequency, global in position, and shared by the star and every control. They lie, from a fixed seed, in an annulus set by each block's own synthesised and primary beams: the field read over is that block's primary beam. The statistic is evaluated at one position at a time, identically at the star and at each control, which is what makes the two comparable. $T_\star$ denotes its value at the star.

Spectra are extracted at the star's apparent position, never from images: for each integration and each native channel the pipeline forms the weighted vector average of the phase-shifted visibilities in total intensity, the mean $I = (XX + YY)/2$ of the two parallel-hand correlations. That average is the dirty-map value at the extraction position – an amplitude, not a deconvolved flux density, so not a point-source measurement where the field holds extended emission – and the correction for its displacement is measured with carriers injected into the same visibilities, not assumed (Appendix A.2). The data are the archive's calibrated deliveries with two exceptions. ALMA's pipeline and online flags were not applied: recalibrating 41 blocks with them takes six of the 33 crossings tested below the trigger, every one unattributed and without a recurrence or in the block failing the quality criterion, and makes the limits of the affected blocks up to 21 per cent too deep, the deepest limit unchanged. And the flux-scaling step was not run on the four blocks calibrated by legacy scripts, which are up to 7.6 per cent too bright. Every delivered channel is searched, with no regridding, deconvolution or edge trimming; a per-integration median along the *frequency* axis removes the continuum, a level and not a shape, and no median is taken along time, so no feature narrow against the band can be removed by it. The delivered channels are smoothed, so flux density times channel width is not yet a radiated power; a response correction computed from the correlator's own channel shape supplies the difference (Appendix A).

One convention holds everywhere in the paper. The search runs on the topocentric sky frequencies ALMA delivered, and every drift rate quoted is the apparent

one at the telescope, which is what the data contain and what a reader can recompute from the archived channel axes. Two steps need the star's own frame: a crossing is compared with the line catalogue, and a repeat epoch is read at the *predicted cell*. That cell is the single channel at a single drift rate where the first epoch's own fit says a carrier should be. Both steps take the star's frame by correcting each block's frequency axis for the Earth's motion at that epoch and then for the star's systemic velocity. That velocity is the catalogued radial velocity from SIMBAD (Wenger et al. 2000), queried on 2026 October 4, and zero where none exists – among the stars with a crossing, for ALMA J153702653−33192492 alone (Section 2). One star departs: toward β Pic the adopted value is $20\,\mathrm{km\,s^{-1}}$, the stellar radial velocity adopted in the ALMA CO analysis of its disk (Matrà et al. 2017). This survey's own blocks reproduce it. It exceeds the catalogued value by about the template-mismatch offset the *Gaia* pipeline documents for stars this hot (Blomme et al. 2023).

Two power quantities carry the main text, both defined in Table 2: the trigger level $P_{\mathrm{trig}} = 4\pi d^2 S\,\Delta\nu$ at distance d, for a carrier that makes $T_\star$ reach $T_{\mathrm{trig}} = 5$ in one native channel of width $\Delta\nu$, and the sensitivity EIRP_{90}, measured by injecting synthetic carriers of known amplitude into calibrated visibilities and recovering them through the unmodified pipeline (§4.1). Every limit quoted in this paper is the EIRP_{90} of its own window, given to two significant figures beside its native channel width, except the bare-threshold comparisons of §1 and Fig. 1(a), which are labelled as such; its error budget is Table 3. The amplitude scale is checked twice: an independent reduction of the same AU Mic visibilities is reproduced to 0.994 ± 0.014, which tests the extraction and not ALMA's flux transfer, and the published quiescent flux density of ϵ Eridani is recovered to 1.10 ± 0.31 (Burton et al. 2022), which measures the scale to 28 per cent. The absolute scale is therefore carried at ALMA's own band-dependent accuracy specification: ±11 per cent for the sample as a whole and ±21 for the 14 Class A windows in Bands 8 and 10, at 492–493 and 849–873 GHz.

The drift range searched is bounded by an acceleration, not by a frequency. Drift rate and line-of-sight acceleration are one quantity in two sets of units,

$$\frac{\dot\nu}{\nu} = \frac{a_{\mathrm{los}}}{c} \qquad \text{and so} \qquad a_{\mathrm{los}} = \frac{c\,\dot\nu}{\nu}, \tag{2}$$

so a ceiling expressed as $\dot\nu/\nu$ is the same physical limit in every band. The default ceiling is 12.0–$13.3\,\mathrm{Hz\,s^{-1}\,GHz^{-1}}$, the acceleration of a transmitter orbiting a solar-type star at a few hundredths of an au. That is the Mason et al. (2025) fiducial with a threefold margin, and 6 per cent of the 200 nHz that Sheikh et al. (2019) recommend for a general search. The ceiling is widened for a confirmed exoplanet host whenever the innermost known planet demands more. As an acceleration it is $3.60\,\mathrm{m\,s^{-2}}$, the median over the Class A windows, rising to $4.00\,\mathrm{m\,s^{-2}}$ on the 5 TRAPPIST–1 windows; the grid is uniform in $\dot\nu$, fine enough that a carrier midway between two trials loses no more than 0.26 of a native channel over the longest track, and runs from the negative ceiling to the first trial at or beyond the positive one,

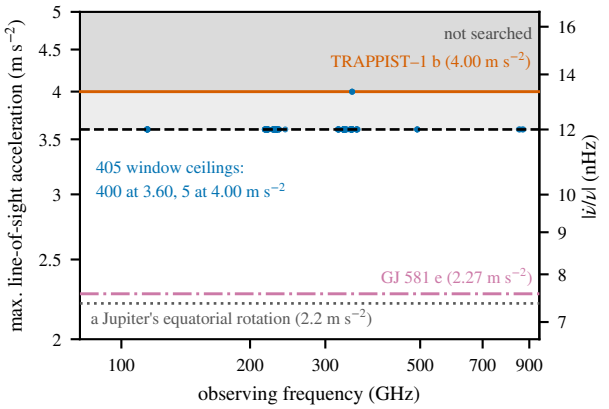


FIG. 3.— What the drift search reaches, as a physical selection function: the drift ceiling of each of the 405 Class A windows (points), by Equation 2, against observing frequency, with the unsearched region shaded and the same quantity as a fractional drift rate on the right. Horizontal lines are the reference accelerations of §3.1.

the largest measured $4.01\,\mathrm{m\,s^{-2}}$ (Appendix D.6). **The survey does not constrain a transmitter whose apparent acceleration exceeds that range**, and three cases put one there. The first is a deliberately chirped carrier, which is in any case not distinguished from an accelerating platform. The second is a transmitter not bound to a known planet. The third is the sum of orbital and rotational acceleration, which share the one budget, so a transmitter on a rapidly rotating planet can exceed the ceiling on its spin alone. β Pictoris b, the fastest rotation measured for any planetary-mass object and orbiting one of the stars searched here, corresponds to 19.4 nHz (Sheikh et al. 2019). For the most demanding known case, transiting TRAPPIST-1 b, 29 per cent of the orbital phase lies above the default ceiling. Its maximum line-of-sight acceleration equals the widened grid used on that star's own windows to within the ±3.1 per cent its host's mass and orbit are known to. The fraction of phase above that grid is therefore anywhere between nothing and a sixth (Fig. 3).

Only part of the archive can carry a drift search at all: the maximum searched drift must cross of order one channel or more over the on-source track, $\eta_{\mathrm{drift}} = |\dot\nu_{\mathrm{max}}|\tau_{\mathrm{track}}/\Delta\nu_{\mathrm{ch}} \gtrsim 1$. 391 of the 405 Class A windows satisfy it and 14 fall below, in every case because the track is short and not because the channel is coarse; the released per-window catalogue carries η_{drift} for every window. 38 of the 1258 coarse windows meet the criterion and stay in Class B, because meeting it is necessary for drift discrimination and not sufficient (Table 2); all of their systems already carry fine-channel coverage and none of the 38 holds a crossing.

3.2. *From a crossing to a confirmed signal*

Every window passes through the same three steps in the same order – the trigger, the line attribution and the recurrence test – and what comes out of them is a *crossing*, an *unattributed crossing* and, had one survived, a *confirmed signal*, each defined in Table 2 and used in no other sense. None of the three is a significance test. Every count reported below is read from the ledger, which lists every crossing the search produced, including those in the archival-tail blocks and in the quality-failing block. The counts are of primary crossings, one to a window;

the 21 secondary crossings of §4.3 pass through the same steps, three reaching the second and none the third.

The trigger. The level on $T_\star$ carries no tail probability, since T is not shown to follow any standard distribution under the null, and a window holds between 112 and 6.87×10^6 channel×drift cells, so one level buys very different protection in a coarse single-pointing window and in a finely channelised one on a long track. Every cell that reaches it is recorded, not only the largest in each window, because a carrier is masked as effectively by a noise excursion as by a line.

The background. Because a fixed threshold is not a fixed false-alarm rate, the number of crossings expected by chance is computed window by window from the exceedance rate at that window's own control positions, and the reserved blocks test the one assumption this makes, that a star behaves like a control position in its own field (§4.3.1); a crossing in them is reported regardless, since a transmitter there would matter whatever its calibration role. The comparison needs each block's control ensemble to behave like noise, and one block, A002_Xc7fa6f_X28a8 toward BD+05 1668, fails the criterion of Appendix D.2. It holds 4 of the 56 crossings, the survey's entire Class B crossing population, all in coarse-channel windows, so the Class A results are identical with it and without it, and the quality-passing data hold 52 crossings, all of them Class A. 1 of those lies in a window the same diagnostic flags inside a passing block; it is counted, and withheld from the amplitude-dependent tests where each is made. The 50 in the primary census are the baseline every expectation is set against, the two in archival-tail blocks enter none, and the failing block's 4 are reported in the ledger and not as crossings of the same standing (Appendix D). Three corrections precede the statistic: the extraction field is chosen so that the star lies inside the delivered primary-beam field and not merely inside the quoted field of view; the stellar position is propagated to the observation epoch for proper motion, though not for the annual parallax or for orbital motion, whose omissions are corrected window by window in §4.1 and bounded for every component of a multiple in Appendix A.2; and on-source time is accounted per integration from the delivered exposure records and not from the scan intentions.

Two diagnostics attach to the second step, and each says which crossing deserves attention first without deciding anything. The first ranks $T_\star$ against the same statistic at the window's 512 control positions, none of which sits on the star; its lowest value, $1/(512+1)$, means the star beat every control. It is a ranking filter and not a significance test, because the stellar and control positions are not demonstrably exchangeable: the annulus responds to whatever it surrounds, and no parameter-free correction for its geometry restores an exchangeable statistic (Appendix B). Weighted star by star, because the five largest stars of the reserved set carry 108 of its 315 windows, the census median rank is 0.49 (0.47–0.51) and the reserved blocks' 0.45 (0.42–0.55), both consistent with the one half exchangeability implies. The rank responds to compactness and not to origin: genuine carbon monoxide toward β Pictoris ranks below its own control ring, because the resolved disc lifts the ring with the star. The second diagnostic, a visibility-domain fit, asks whether a crossing's emission is at the stellar coordinates, but it cannot discriminate at the trigger either, because at that level a noise maximum at the star fits as a point source there (§4.5). Neither gates a disposition, so neither is charged to the sensitivity. Throughout, p denotes a genuine significance level, and no probability computed against the rank's own null is called one.

The line attribution is a single database query, made on 2026 October 6 and repeatable as it stands, and its $\pm 50\,\mathrm{km\,s^{-1}}$ half-width follows the Keplerian line-of-sight velocities of gas surviving beyond a few au with margin for the systemic velocity; a crossing inside it, in the star's own rest frame, is *attributed* where the species is one that such systems are known to carry, and a further three crossings coincide with a species no debris disc has been shown to carry, whose offsets are reported and which are counted as unattributed (Appendix C). Halving the half-width leaves the split unchanged: 18 attributed and 38 unattributed at $\pm 25\,\mathrm{km\,s^{-1}}$, against 18 and 38 at $\pm 50\,\mathrm{km\,s^{-1}}$. A coincidence is a statement about a catalogue and not a property of a frequency, so an attribution is not a disposal: all 56 crossings go on to the recurrence test whichever side of the mask they fall, and the price of the mask is that the survey constrains nothing inside it.

The recurrence test. One reason covers both halves of *separated*: blocks of one session share calibrators, atmosphere, correlator setup and interference environment, and the residual of §4.3.2 is what that can produce, which holds for two blocks of one session however far apart they were executed – 31 pairs of blocks here are of one session and more than a day apart, the widest by 18.02 d. So a recovery in any block of the discovery's own session is a *within-session repeat* whatever the interval, and no number of them makes a confirmation; this is also the rule the confirmability census of §5.1 scores the archive on. What a confirmation requires, and the two readings that test it, are defined in Table 2. It is a weak criterion, and the archive's uneven temporal sampling is why: only part of the sample has epochs separated well enough to apply it at all.

That the chain finds what it is meant to find is shown twice over, by the injected carriers of §4.1 and by β Pictoris CO – a real feature at a known stellar position, recovered and localised in two transitions and 10 execution blocks (§4.4).

4. RESULTS

The primary experiment is the Class A search for a drifting, unresolved spectral carrier: 405 fine-channel windows toward 61 systems. Of those windows, 391 are channelised finely enough on the criterion of §3.1 to tell a drifting carrier from a stationary one. The Class B search for an unresolved spectral excess is secondary and carries no drift discrimination: 1258 coarse windows. Together they are the 1663 windows of the primary census. No event satisfied the confirmation criterion set out in §3.2.

4.1. *Sensitivity: the power at which a carrier would have crossed the threshold*

The median system, on its best window, would have produced a threshold crossing from a carrier of 1.8×10^{15} *W.* Each window's limit is taken over the channels the attribution mask leaves, in the star's own rest frame

TABLE 2
THE QUANTITIES AND THE THREE WORDS THIS PAPER USES, DEFINED ONCE. EVERY TERM BELOW IS USED IN THIS SENSE THROUGHOUT AND IN NO OTHER. THE DISTINCTION THE FIRST THREE ROWS DRAW IS THE ONE THE LIMITS OF §4.1 TURN ON: A CROSSING IS A SINGLE-EPOCH EVENT, AND A CONFIRMED SIGNAL NEEDS A SECOND AND SEPARATED EPOCH THAT NO SENSITIVITY SUPPLIES.

Term	Definition
Crossing	A channel × drift cell at the stellar position reaching $T_\star \geq 5$ in one execution block; neighbouring channels above the trigger belong to one crossing. The largest in a window is its *primary* crossing and any other a *secondary* one, and because the largest cell is the window's own statistic a secondary crossing exists only where a primary one does. An event requiring investigation; no claimed significance.
Unattributed crossing	A crossing lying more than $50\,\mathrm{km\,s^{-1}}$ in the star's frame from every transition of the line catalogue. That is a reproducible frequency coincidence; which coincidences are astrophysically plausible is a separate judgement, stated with each one. A target for follow-up; not a detection.
Confirmed signal	The same carrier found again in a *separated* block – a different observing session, more than 1 day from the discovery – at a frequency one bounded trajectory inside the drift grid joins to the discovery. Two readings test that, and they are not the same test. The *line-recurrence check* is the operational one: it reads the single channel and drift the discovery predicts in the star's frame, so it recovers only a carrier that comes back to that cell, within ±0.08 to $\pm0.63\,\mathrm{km\,s^{-1}}$ – which is what a spectral line does and an accelerating carrier need not. It is applied to every crossing with a covering repeat block, attributed or not, and a null there is a statement about attribution and never an exclusion; 13 line-attributed crossings pass it, this stage's positive control (§4.6). The *trajectory comparison* searches the $\pm833\,\mathrm{km\,s^{-1}}$ about that cell and asks whether one bounded trajectory joins two crossings; it is the reading that bears on a drifting carrier. It separates 1 of the 13 pairs and has never accepted a drifting carrier, so a pair it cannot separate is *open*: neither a confirmation nor an exclusion (Table 5). Neither reading exists unless the archive covered the frequency again in a separated block, and for 21 Class A systems it never did, so no sensitivity makes a confirmed signal possible there.
P_{trig}	The power of a carrier that makes $T_\star$ reach the trigger in one native channel. An internal pipeline level: not a sensitivity, not a significance.
EIRP_{90}	The power at which nine injected carriers in ten produce a crossing at the star: a measured single-epoch trigger completeness, not a detection probability. Measured for the 49 injected Class A windows; for the other 356 an estimate from the class-average calibration, interpolated within the injected configurations for 294 and extrapolated beyond them for 62 (§4.1). Nominal in one respect: it excludes atmospheric decorrelation, which Table F1 gives system by system.
Class A	The primary experiment, defined by channel width: 405 fine-channel windows, median 488.3 kHz, toward 61 systems over 47.7 GHz of union bandwidth, 391 of which satisfy $\eta_{\mathrm{drift}} \gtrsim 1$ (§3.1).
Class B	The secondary experiment: 1258 coarse-channel windows, mode 15.625 MHz, constraining a persistent unresolved excess. Never a carrier search: in 1220 of them no searched drift leaves one channel, and in the other 38 the whole searched range moves a carrier 1.02–1.35 channels, which registers one but does not discriminate it (§3.1).
The sample	Neither volume-limited nor representative of the solar neighbourhood: the stars ALMA happened to observe for other reasons (§2).

(§3.2): the finest windows were tuned to CO, a 15.26 kHz window spans 53.9 MHz against a mask 76.9 MHz wide, and 15 of the 405 Class A windows, the two finest among them, lie wholly inside it and constrain nothing. Across the 60 systems left with a constrained window the limits run from 8.7×10^{13} to 3.7×10^{16} W (Fig. 5), 24 of them reach 10^{15} W and the median Class A *window* reaches 3.1×10^{15} W; for HIP 10679 every fine window is inside the mask and no limit is quoted. In 44 blocks whose in-beam fields were recombined after the limits were computed, the search ran on the recombined data, so the quoted limits there are conservative: for the 8 systems whose adopted limit such a block sets, by a median factor of 2.1 and at most 5.7. Over the 391 windows in which the searched drift crosses a channel the search covers 44.4 GHz toward 58 systems, and restricting each system to them gives a median 14.1 per cent shallower.

Carriers were injected at a random sub-channel phase and a random drift from each window's own grid and recovered through the unchanged pipeline, through the trigger and the localisation at the stellar position inside one execution block. Nine in ten come back at $3.03\,P_{\mathrm{trig}}$ in Class A, P_{trig} being the bare 5σ of the window itself, and that power is EIRP_{90}; it does not depend on the brightness of the control ring (Fig. 4), and the correlator's spectral response is carried by the injected tones and so sits inside it (Appendix A). The 512-position spatial rank is not charged against it: that would cost a further factor 1.31, and in 3 bright-ring windows it could not be met at any injected power. Class B carries its own factor, $4.17\,P_{\mathrm{trig}}$ from 5 936 carriers injected into 35 coarse windows of 14 execution blocks, which sets the limits for Barnard's Star, Wolf 359 and Proxima. The calibration interval of the class factor, ×0.89–×1.11 (Table 3), applies to the median and the counts and not to any one window's limit: the median system's limit carries it, 1.6×10^{15} to 2.0×10^{15} W, so the second figure of that median is set by the calibration and not by the depth of any observation.

Every limit quoted here is nominal in one respect: it excludes atmospheric decorrelation, which a carrier injected into calibrated visibilities does not suffer and a real one does. Measured block by block from the phase rms the observatory records on each block's phase calibrator (Cortes et al. 2026, §§10.4.11, 11.3.1) through the standard phase-scatter relation (Richards et al. 2022, Eq. 3), it leaves the median system's best window optimistic by 3 per cent – 1 per cent for the median window of the census, 8 per cent above 400 GHz, and 200 per cent for HD 172555. It is stated and not folded in because it is

available for 279 blocks of 407 only, the rest measurably the harder ones, and folding it in where it is known would leave limits neither corrected nor uncorrected. Table F1 gives it system by system (Appendix F). Applied there where a record allows it, it moves the median limit above by 2.6 per cent – inside the two figures these powers are quoted to – and takes the 24 systems reaching 10^{15} W to 22. No system's best window lies above 400 GHz, so the high-frequency figure touches none of the limits quoted here.

The primary-beam response at the star's own offset, the annual parallax the phase centre omits and the smearing of a carrier within one integration are divided out window by window, each bounded in Table 3 and defined in Appendices A and A.2. *The net bias on one limit is therefore of either sign*: the injected profile's 8 per cent shortfall against the exact channel response makes a limit conservative and the decorrelation makes it optimistic. In the median window the first is larger and the quoted limit is conservative by 7 per cent; at the ninetieth percentile of the systems' own best windows the second is, and the limit is optimistic by 18 per cent net of that shortfall – 27 per cent before it – rising to 178 per cent net and 200 gross for HD 172555.

Carriers were injected directly into 49 of the 405 Class A windows, 12 per cent, and the result transferred to the remaining 356. That transfer is the dominant caveat on everything above. The injected windows are those of whole execution blocks drawn in a random order and balanced by band and array, and over 6 properties they span the central eight-tenths of the population, five of the six agreeing in the median to 6.3 per cent and the sixth, the noise level, ×1.7 noisier (Appendix F). Only the ninety per cent point in units of the window's own trigger power is carried across; everything that turns it into a power is that window's own measurement. Across the injected windows that point ranges from ×0.80 to ×1.58 of the adopted factor, with 41 of the 49 within a tenth of it. *That dispersion, and not the calibration interval above, is what one uninjected window's own limit is uncertain by.* On the five properties that set the reading – band, channel width, array configuration, integration time and block duration – 294 of the 356 transferred windows lie inside the injected range on every one, and 62 outside it on at least one, where the range just quoted is not an empirical bound. 8 systems take their headline limit from one of them, and none is among the 24 systems reaching 10^{15} W. The best window was itself injected into for 6 of the 60 systems, marked as measured in Table F1; for the other 54 the headline limit is an estimate carrying that dispersion. *The median quoted above is therefore a calibration-transferred estimate and not a measured sensitivity.*

A carrier that is not always radiating is penalised in proportion: over 3 888 undrifted trials at duty cycles down to 10 per cent the recovered amplitude followed the duty cycle to 1 per cent, so a transmitter radiating a tenth of the time needs 10× the power quoted above, the joint drifting and intermittent case being unmeasured.

4.2. *The nearest stars: Barnard's Star and Wolf 359*

Barnard's Star is the second-nearest stellar system and Wolf 359 the third if brown dwarfs are excluded. To our knowledge these are the first millimetre or submillimetre

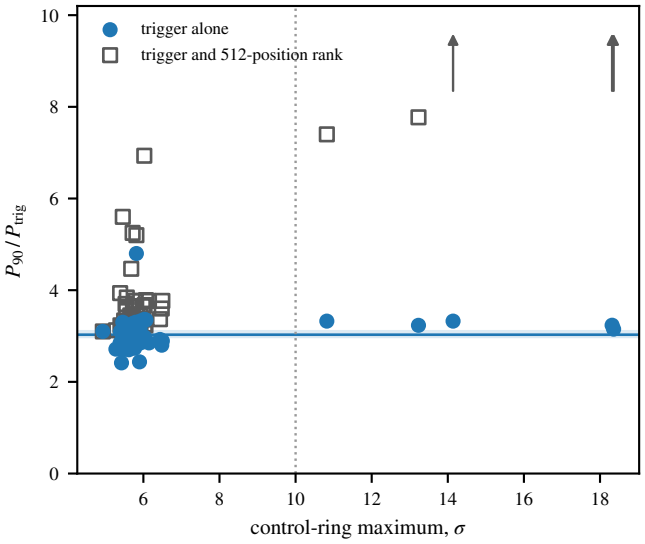


FIG. 4.— Why one completeness factor serves the whole class. The 49 directly injected Class A windows at the power recovering nine carriers in ten, each against its own 512-position ring maximum: no trend with the ring. Filled, the criterion adopted here, pooled value as line and band; open, the rank charged too, arrows where no power meets it; dotted, a resolved ring.

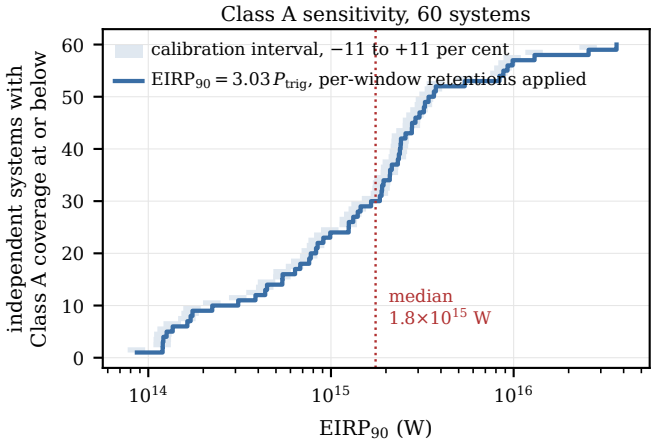


FIG. 5.— **Single-epoch trigger completeness of the carrier search**: the cumulative number of systems whose best window reaches a given EIRP_{90}, each window's own retentions applied and atmospheric decorrelation excluded, the calibration interval of Table 3 shaded. Confirmation needs a separated repeat, which no power buys and which 21 of the 61 systems lack (Fig. 2b; §5.1).

technosignature limits for either. Radio searches of the nearest stars have worked at centimetre wavelengths (Enriquez et al. 2017; Price et al. 2020; Margot et al. 2023). Both are non-detections, and *both limits come from the secondary Class B experiment and not from the carrier search*: the archive holds only coarse Band 6 windows toward these stars. On the EIRP_{90} convention of §4.1 Barnard's Star reaches 2.7×10^{14}–4.0×10^{14} W over 12 windows and Wolf 359 3.1×10^{14}–4.5×10^{14} W over 12.

4.3. *Threshold crossings and how each was disposed of*

Of the windows the search covers – the 1663 of the primary census and the archival-tail blocks beside them – 56 hold a cell reaching $T \geq 5$ at the stellar position. 4 lie in the one execution block that fails the quality criterion of §4.3.2, so the population that characterises the search

TABLE 3
WHAT SETS $EIRP_{90}$, AND IN WHICH DIRECTION. THE CALIBRATION INTERVAL APPLIES TO THE CLASS-AVERAGE FACTOR AND SO TO ANY STATISTIC OVER MANY WINDOWS; THE SPREAD BETWEEN WINDOWS IS AN OBSERVED DISPERSION AND IS THE STATEMENT ABOUT ONE WINDOW; THE SIGNED TERMS ARE BIASES AND ARE NOT FOLDED INTO AN ERROR BAR (§4.1).

	value	what it does to the limit
Class-average completeness	×3.03	nine carriers in ten recovered at this multiple of a window's own trigger power. *Measured* in the 49 Class A windows injected directly and *estimated*, at this factor, in the other 356
Uncertainty on that calibration (per cent), combined in quadrature		
The factor itself	±2	sampling error of the pooled value over the 49 injected windows
Absolute flux scale	±10.2	the observatory's own band-dependent specification (ALMA Proposer's Guide (Privon et al. 2026), with the Technical Handbook's quality-assurance criterion (Cortes et al. 2026) at Band 8, where the two differ): ±5 in Bands 1–5; ±10 in Bands 6 and 7; ±20 in Bands 8–10, weighted over this sample's bands
Distance and visibility scale	±4.5	the Gaia parallaxes, which enter as d^2, and the block-to-block visibility scale
Combined	$-11/+11$	×0.89–×1.11 on the factor, and so on the median limit and on the system counts; ±21 for the 14 Class A windows in Bands 8 and 10, at 492–493 and 849–873 GHz
Spread between windows. An observed dispersion, not an error bar.		
Window to window	×0.80–×1.58	the 49 injected windows resolve 2.41–4.80 $P_{\rm trig}$ individually, 41 of them within a tenth of the pooled value and the second largest at ×1.11. This, and not the interval above, is what an estimated limit is uncertain by; it is wider, and it applies to 54 of the 60 systems
Applied window by window where a record allows it and bounded where none does, and so not combined with the above.		
Intra-integration smearing	×1.00–×1.05	a carrier sweeps through part of a channel within one integration and the injected ones do not: $\eta_{\rm smear}$ at the drift that sets a ninety per cent point, from each window's own integration length where it is recorded (1054 windows) and from the longest delivered anywhere where it is not (609)
Primary beam	×1.00–×4.3	the star is not at the pointing centre, so each limit is divided by the response at its own offset, on the full width the observatory documents. Nothing in the median window; the 12 windows furthest off-axis carry the departure of that form from a blocked circular aperture, up to 6.2 per cent, as a one-sided systematic
Annual parallax	×1.00–×2.06	the phase centre omits it, so each limit is divided by the amplitude a compact source retains at its own displacement, a median 0.13 per cent on the 1650 windows carrying a parallax record; on the other 13 the loss is bounded (Appendix A.2)
Signed terms. A bias is not an error bar and is not added to one.		
Channel response	×2.29	one over the share of a channel-confined carrier the peak channel keeps. Carried by the injections, so already inside the completeness above, which is this factor times a margin of ×1.325 over the nominal trigger
its residual	+8	the injected profile falls short of the exact response: *conservative*
Decorrelation	$+0/+200$	measured block by block from the phase rms the observatory records on each block's phase calibrator, which an injected carrier does not suffer and a real one does. One-sided, and a floor. A median +3 on the systems' best windows, +8 above 400 GHz, and more than +20 in 5 per cent of windows

is the 52 in quality-passing data and those 4 are reported beside it. The molecular-line mask is $\pm 50\,{\rm km\,s^{-1}}$ wide in each star's own rest frame and places 18 of the 52 within that distance of a catalogued transition. A further three fall on transitions of a species no debris disc has been shown to carry. The mask does not hold those transitions, so they stay unattributed, with their offsets reported (Appendix C). That leaves 18 crossings attributed and 34 unattributed. The attributions are kinematic and not an accident of bandwidth. In the frame the correlator delivered, the crossings with a measured peak frequency crowd onto the transitions these setups were tuned to, far more tightly than a free landing on their own grids would put them (Appendix E). Of the unattributed, 2 outrank all 512 control positions of their own window, and neither of those two is present again at the cell its own discovery predicts. No crossing survives the chain, and this survey reports no detection; the crossing ledger, which accompanies this paper as data, lists every one.

Anything bright in a window hides a weaker carrier behind it, so the weaker cells were sought too. The de-drifted statistic is retained in every channel. Behind a window's tabulated crossing, its largest cell, weaker above-trigger cells can therefore be read as well, neighbouring channels counting as one. There are twenty-one of them, of which three are unattributed and none reaches the trigger in any of the covering blocks the recurrence test reads. Cells are counted in the controls too. Outside the mask the star holds slightly more of them than a control position at every level below the trigger, with no step where a carrier search would put one (Appendix D, §4.3.1).

4.3.1. *The unattributed count is what the control positions predict*

The trigger is a threshold on a statistic and not a false-alarm rate – the searched channel × drift cells per window span 4.8 decades – so the chance count is measured window by window over the 405 Class A windows of the census, with one condition on both sides of the comparison: a cell at the stellar position that reaches the trigger and that the line mask does not attribute (Appendix D gives the method). These fields' own control positions predict 24.8 unattributed Class A crossings and 31 are found. A count that large or larger arises by chance with probability 0.14 – a probability conditional on the control positions being a model of the stellar position, the one assumption the searched data cannot test. Calibrated instead at the stellar-to-control ratio 1.104 measured just below the trigger, as a sensitivity test, the expectation is 27.3 at $P = 0.27$, and the whole-beam conclusion of no significant excess is unchanged under it. A calibration error varying with frequency would raise the star and not its continuum-free controls, but the least demanding window would need 21.4 per cent of its star's continuum to reach the trigger, and the mild excess below it runs the wrong way with continuum brightness, 1.01 against 1.06 over the 350 windows with a photospheric estimate.

Both sides are read over the whole primary beam §3.1 defines, the statistic Equation 1 specifies; a narrower field drops the integrations whose pointing falls outside it and

with them on-source time and real signal. Read over such a field on both sides the same comparison gives 35 against 24.1 at $P = 0.027$, reported beside the first and not in place of it. Both geometries are applied uniformly to every window and to both sides, but neither was nominated as the primary before the counts were compared, and the choice between them was made after the counts had been seen. The whole beam shows no significant excess and the narrower field a nominal one; corrected for the 2 readings tested the smaller probability is 0.05, so neither departs from chance at the 5 per cent level – a correction that assumes only these 2 choices were made, and so does not settle the disagreement. The interpretation therefore depends to some extent on the extraction geometry, and neither reading establishes a population of non-astrophysical transmitters.

The observed count is compatible with a background estimated empirically from these same fields, which is not a null distribution established independently of the model. What the comparison could not have rejected is 17 further crossings against those control positions, 22 on the narrower reading – conditional on that model, a count of crossings and not of transmitting systems, since several crossings could come from one source, and not an occurrence rate (Appendix D). Each crossing is tested individually by the recurrence test of §4.6.

4.3.2. *One execution block accounts for four of the crossings*

4 crossings belong to BD+05 1668 (GJ 273, Luyten's Star), all of them in one ACA execution block, `A002_Xc7fa6f_X28a8`, whose whole field is above threshold. Of its 512 control positions, 512 exceed $T = 5$, where the median of a window's own control ensemble is 2.74 across the survey. Its single-integration noise is as predicted, but it does not integrate down: a jackknife on the channel mean stands 6.2 times above thermal while neighbouring channels stay uncorrelated. That describes a residual persisting from one integration to the next and differing from channel to channel (Appendix D.2). Equation 1 divides by the noise of the stacked spectrum, which falls as the root of the integration count. A residual that does not average down therefore accumulates along a drift track faster than the denominator grows, and it does so at every position in the field. *The field is inflated and the star is not*: the 24 sibling executions of the same field, array and setup produce no crossing, and in all four a control stands above the star, 1–421 of the 512. The criterion fails this block and no other at any threshold between it and the next most extreme block, so its 4 crossings are reported apart from the 52 and Appendix E carries what the block might instead be. Milder cases are not unique to it: read window by window the same diagnostic flags 5 windows in 2 blocks, and the 1 flagged crossing outside this block – HD 48370 – is withheld from the amplitude-dependent tests for it too.

4.3.3. *Stellar flares are excluded by a broadband prediction*

Known flare stars are in the sample, but a millimetre flare raises every channel of a window alike. One able to produce a single-channel crossing at the trigger must therefore produce a $5\sqrt{N}\sigma$ continuum event in the same integrations. With N the window's own channel count, 111–3 769 across the windows holding a crossing, that prediction runs 53–307σ. The observed continuum statistic has a median of -0.29 over the 48 crossings with a recoverable measurement, and $|Z| < 3$ in 45 of them. A flare origin in fact needs more than that, because the per-integration median along frequency (§3.1) removes a flat continuum before the search. Only a flare's departure from flatness across the band can reach a single channel. The prediction is therefore a floor on what such an origin requires and the contribution measured below is a bound on what one supplies. One block settles it: AU Mic `A002_Xa42f75_X319` holds both a crossing ($T_\star = 5.27$) and the millimetre flare of MacGregor et al. (2020). That flare contributes 0.17σ to the single-channel statistic. It raises the continuum at $Z = 10.3$, while the crossing's own de-drifted track reaches $Z_{\max} = 2.1$.

4.4. *β Pictoris: an astrophysical positive control, and the obstacle it identifies*

A search that finds nothing must show that it could have found something, and the survey's strongest features show exactly that. The pipeline recovers the β Pictoris circumstellar carbon monoxide without being told the disc is there. It does so in two rotational transitions and in 10 execution blocks spanning 8.4 yr, drawn from 6 observing sessions, the closest two 25 d apart. In 9 of the blocks the star also outranks every control, but the identification is kinematic and does not rest on those statistics. Carried through the frame chain to the barycentre at each block's own epoch and then to the star's rest frame, all 10 crossings lie within $0.5\,\mathrm{km\,s^{-1}}$ of the stellar systemic velocity. They lie in the two lowest CO transitions of a star with mapped circumstellar CO (Dent et al. 2014; Matrà et al. 2017). The Band 3 peaks agree across blocks to 1.00 channel once each block's own tuning is removed. The alternative reading requires a transmitter placed by coincidence at matched offsets from two CO transitions of one star. It is also the control that shows the recurrence test of §4.6 working in the positive direction. Every flagged feature is recovered in another execution block and each transition in a block of another session as well, the screened Band 6 window returning $T_\star = 8.95$. A non-recurrence elsewhere means nothing unless the same machinery recovers a repeat where one exists. *One of those blocks is a counterexample, and it is the more useful half of the control.* In block `X7116f1_X1785`, Band 6, the control ring reaches 14.13 against 9.68 at the star. Several of the 512 control positions therefore exceed the stellar statistic, and the spatial rank disfavours a window holding carbon monoxide known to be real and centred on the star. The disc is resolved on the scale of the primary beam in that configuration. That is the geometry the statistic is built to reject, so the rank tests compactness and not authenticity, and on a real extended source it errs conservatively (Fig. B1). Extended emission is in fact identified by the *level* of the control ring and not by the star's rank against it. That is why the window-level flag of Appendix D.2 is defined on the ring's median. Appendix C gives the complementary case, HD 48370, whose crossing matches the prominent CO(2→1) that Cataldi et al. (2023) report toward the source and flag as possible cloud contamination. Its control ring rises with the star, as emission filling the primary beam does. The mask that removes

this emission covers 11 species and costs 5.0 per cent of the Class A union bandwidth, and it is nevertheless the single most productive test in the chain (Appendix C). §5.1 draws the lesson.

4.5. *The two crossings that lead their own control fields*

Two crossings reach a stellar statistic above every one of the 512 control positions in their own window, and neither is a detection: 61 Vir at 346.3232 GHz and CP−72 2713 at 344.2965 GHz, both in Band 7 and in the star's own rest frame (the delivered frequencies are in the ledger), tabulated in Table 6 and the first drawn in Fig. 6. Their control rank says which crossing was examined first and disposes of nothing (§3): were the stellar and control positions exchangeable, the star would lead all 512 with probability 1/513, so 0.07 of the 34 unattributed crossings would do so by chance and two or more with probability 0.002, against 0 of 6 in the reserved blocks and 0 of 6 beyond 40 pc, and the positions are not exchangeable (§3.2). CP−72 2713's crossing lies $+1236\,\mathrm{km\,s^{-1}}$ from CS(7–6), a note and not a disposition (Appendix C). A visibility-domain fit, made for 51 of the 56 crossings and consistent with a point source at the star for 41, excludes neither, injected sources in the bin these two occupy returning $\mathrm{Re}/\sigma = 6.58 \pm 1.97$; of the 512 control positions, 8 retain a dynamic spectrum and 12 are reachable in the visibilities. Neither crossing is present when the same stellar-frame frequency is observed again: the nearest covering block, 1.97 and 2.05 hours from each discovery, bounds a returning carrier below 0.95 and 0.71 of the discovery amplitude, and every covering block combined deepens that to 0.56 and 0.66 over 3 and 2 days. Either reading covers half a channel in the star's frame, ± 0.211 and $\pm 0.213\,\mathrm{km\,s^{-1}}$, which each carrier's own fitted drift would cross in 77 and 119 seconds, so the test asks for a return to that frequency and not a stability at it (§4.6); T_{rep} is one cell at one trial and may be negative, and the recovery chance in brackets, at the discovery amplitude, is biased high. At the faintest amplitude consistent with each crossing having been found the criterion would have recovered a persisting carrier with calculated probability 1.00 and 0.66, and injected carriers measure 0.99 and 0.63, so for CP−72 2713 the silence is not an exclusion. The readings behind each bound – the largest single epoch, the maximum over every drift and the value over the matched registration tolerance – are in `candidates/repeat_epoch_readings.csv` of the data archive (White & Dey 2026).

4.6. *Recurrence, applied across the unattributed crossings*

Because a transmitter in orbit changes its line-of-sight velocity between epochs, the broad test is whether anything above the trigger is present again anywhere in a covering window, at every drift of the grid. That window is $\pm 833\,\mathrm{km\,s^{-1}}$ wide in the median case, wider than the $\pm 316\,\mathrm{km\,s^{-1}}$ an orbit at the grid's $3.6\,\mathrm{m\,s^{-2}}$ ceiling can produce about any star here. 17 of the 212 distinct covering windows hold a statistic above the trigger, against the 11.3 chance predicts ($P = 0.05$, or 10.4 at $P = 0.03$ when the rate comes from each window's own control positions instead of its cell count; Appendix D) – a marginal excess, and every one of those windows holds a crossing the ledger already lists, so this comparison rereads the trigger-level excess of §4.3.1 rather than adding a recurrence to it. 10 of the 17 are toward HD 14055 and ALMA J153702653-33192492 (§4.7); over the remaining 161 windows the reading is 7 against 4.8 at $P = 0.19$ (Appendix D.5). No recurrence is established here, and none is excluded.

Table 5 asks of the 13 pairs of crossings inside that interval, across 5 stars, whether one transmitter could have produced both (Appendix D.5). Constant line-of-sight acceleration means $\dot{\nu}_1 = \dot{\nu}_2$, and the two fitted rates differ by more than the grid's own rate resolution in 13 of the 13 pairs. In 7 of them the epochs are close enough together that the offset such a trajectory predicts is tighter than the window searched about the cell, and those 7 fail a drift changing at a constant rate as well. The weaker and safer question, whether any trajectory inside the searched grid joins the two, separates 1 pair, HD 14055's two crossings needing 25 times the grid's ceiling between them; the other 12, 10 *open* and 2 *edge-limited*, are neither a confirmation nor an exclusion, since the comparison can reject a drifting carrier and cannot accept one.

A third reading is a check on line attribution, and not a test of persistence: the single cell each discovery predicts at the repeat epoch, maximised over nothing, which is right for an emitter at rest in the star's frame, as a line is, and wrong for a carrier: the fitted line-of-sight accelerations of the 33 run from 0.48 to $3.64\,\mathrm{m\,s^{-2}}$ in the observatory's frame and from 0.45 to 3.66 in the star's own – the two frames differ by the observatory's own acceleration and so by at most $0.035\,\mathrm{m\,s^{-2}}$ – and a drift that size carries a carrier out of the half-channel the reading covers in 38 to 471 seconds, a median of 111. The shortest interval any of them has between its two epochs is 23 times the longest of those residences. It is therefore reported here as a *line-recurrence check*, and no crossing is called excluded on it. Nor could it be extended to a carrier by carrying that carrier's own fitted drift forward: the half grid step the drift is resolved to holds the predicted cell inside half a channel for a median 2.0 h (0.3–2.4), which 23 of the 33 exceed before their nearest repeat, and it displaces the cell by a median 5 channels there and 648 over the longest baseline. It works in that direction: 13 of the 15 crossings attributed to carbon monoxide recur in another session, the strongest recovery of each 39–1984 d away – 10 in the disc of β Pic and 3 in interstellar gas toward HD 285968 (Appendix C) – but every disc recovery has $T_\star \geq 8.0$, and the 2 not recovered are the two faintest, both at $T_\star = 5.41$. On the 37 unattributed crossings whose predicted cell could be read it returns nothing: none is present at its predicted cell, the largest single-block reading at the carried drift being 2.48, and 4.71 the largest at any drift of the grid, against a trigger of 5; the remaining 1 of the 38, toward HD 14055 at 331.1559 GHz, retains no discovery spectrum, so there is no amplitude to carry to a repeat epoch and no reading of its own. Over the 33 crossings whose repeat coverage carries an amplitude bound, a carrier returning to within half a channel is bounded to a median 0.43 of the discovery amplitude and 0.997 at the weak end, where 8 are too shallow to test at the trigger amplitude; no disposition depends on the cell's registration, on the anchor along the discovery track or on the threshold (Appendices B.2

and D.4). No crossing satisfies the confirmation criterion of Table 2; the repeat coverage holds the marginal excess of above-trigger windows given above, concentrated in HD 14055 and ALMA J153702653-33192492; and no reading here bears on a transmitter whose apparent frequency or acceleration changes between epochs by more than these readings accommodate.

4.6.1. *The reserved hold-out and the control beyond 40 pc*

Three sets of execution blocks enter no census statistic (Table A1); each was searched with the same pipeline, every compact-array block on the annulus of its own dish (§3.1), and each is reported here as a search. The *hold-out* is 77 blocks, 315 windows, 36 stars and 233 h on source, set aside by the identifier rule of Appendix B.1 and sharing no execution block with the census. It contains 6 threshold crossings, the strongest at $T_\star = 8.53$, none of them outranking its own control positions. Of the 4 tested for recurrence on the same code path exactly one is present again – a molecular line, so the test works in the positive direction here as well, at $T_{\rm rep} = +6.73$ – while for every other tested crossing the repeat coverage bounds a carrier returning to the predicted cell to 0.42 of the discovery amplitude or less. The *archival extension*'s 463 windows in 115 blocks hold 10 threshold crossings, every one in a fine-channel window. Of these, 3 lie inside the mask in their own star's frame and 7 outside it, against 4.7 predicted by chance (Appendix D). The chance of 7 or more is 0.20. A sample this size could have shown a shortfall and does not, $P(0\,|\,4.7) = 0.009$. The *control beyond 40 pc* is 47 blocks and 198 windows toward 22 stars, 88 h on source, holding 6 crossings, the strongest at $T_\star = 13.2$ and none outranking its own controls; the set is read in the sky frame, supports no limit and shows only that the pipeline behaves the same way on fields beyond the sample. The hold-out crossings are released row by row with the catalogue, on the columns of the crossing ledger.

4.7. *HD 14055 and ALMA J153702653−33192492*

Of the 17 covering windows above the trigger, ten are toward these two objects; Table 4 is every crossing either of them holds.

They are also the two most heavily observed objects in the survey, holding 20 per cent of the Class A census and 51 of the 212 covering windows, so 6.5 of the 11.3 expected exceedances are theirs before anything is looked at. Taken one at a time the two part company. HD 14055 holds 6 against 5.3 expected ($P = 0.45$; the control positions of its own windows predict 5.0): its concentration is its exposure. ALMA J153702653−33192492 holds 4 against 1.1 and 1.0 on the two nulls ($P = 0.024$), a real local excess – though it is the most discrepant of the 19 objects compared and was picked out for being so: chance puts some object that far out in 14 per cent of realisations.

Nothing about HD 14055's fields is unusual: the median of a window's control ring over the median of its own class is 1.04 for its windows, against 1.00 for the survey and 4.1 for the one block the quality criterion rejects (Appendix D.2). Its setups are continuum ones, the nearest masked transition to any window centre 568 km s^{-1} away, and its crossings lie 238–1166 km s^{-1} from CO or ^{13}CO in its own frame. Its six catalogued crossings come from blocks of a single programme taken within 2.1 d of each other, and Table 5's 651 d baselines come from its two archival-tail crossings; the 9 epochs that carry the inverse variance there are follow-up blocks outside the primary census, and on them a carrier persisting at the discovery amplitude is excluded at 11.8 and 12.7σ. Two of its drifts lie on their grid's outermost trial and are lower bounds, $|a_{\rm los}| \geq 3.60$ and ≥ 3.63 m s^{-2} – at the top of the range searched, where §3.1's caveat applies, and the reason two of this star's pairs in Table 5 are edge-limited.

TABLE 4
Every crossing toward HD 14055 and ALMA J153702653−33192492. Rates are in the observatory frame; $\geq$ and $\leq$ mark a fit on the grid's outermost trial.

$\nu_\star$ (GHz)	$T_\star$	$\dot\nu_{\rm obs}$ (Hz s^{-1})	nearest masked transition	$\Delta v_\star$ (km s^{-1})	Δv_{225} (km s^{-1})	Re/σ
HD 14055						
331.155	5.08	≤ -3976	^{13}CO(3–2)	+514	−1	+4.4
331.703	5.28	+1104	^{13}CO(3–2)	+1012	−5	+5.1
331.766	5.58	−3857	^{13}CO(3–2)	+1069	−12	+5.0
344.452	5.01	−2589	CO(3–2)	−1166	+4	–
344.696	5.17	−2426	CO(3–2)	−953	−3	+4.8
345.007	5.18	+1769	CO(3–2)	−684	+2	+4.6
345.430	5.45	+2584	CO(3–2)	−317	−3	–
345.522	5.19	$\geq +4184$	CO(3–2)	−238	+1	+5.0
ALMA J153702653−33192492						
216.204	5.23	−2198	SiO(5–4)	−1245	−5	−0.6
217.727	5.09	−983	$H_2CO(3_{0,3}–2_{0,2})$	−680	+3	+1.8
218.669	5.05	−2023	$H_2CO(3_{2,1}–2_{2,0})$	−125	−6	+4.5
218.892	5.09	+2373	$H_2CO(3_{2,1}–2_{2,0})$	+181	−15	+4.7
230.539	5.41	−19	CO(2–1)	+2	+2	+4.4
347.189	5.01	−2581	SiO(8–7)	−122	−7	+4.6

ALMA J153702653−33192492 is a serendipitous dwarf 5.75 arcsec off the phase centre of fields pointed at the Lupus clouds, in 212 s snapshots. Its CO($2 \to 1$) crossing is the only one of the 14 sitting on a line the observers tuned to: window centred 16 km s^{-1} from CO, fitted drift 0.6 of a grid step from zero where every other is 15–71, and a control ring reaching $T_\star = 13.4$ against 5.4 at the star – cloud emission filling the field. The star has no measured systemic velocity and the frame chain takes zero (§3.1), so the attribution of all six was recomputed over ±50 km s^{-1}, wider than any systemic velocity adopted here. One changes: the CO crossing leaves the mask above $v_{\rm sys} = +48$ km s^{-1}, where its reading in the local standard of rest, 0.5 km s^{-1} from a catalogued cloud line, holds anyway (Appendix C). The other five never enter the mask at any velocity of the sweep, the closest approach of any of them to a masked transition being 72 km s^{-1}.

What is left is four windows where 1.1 were expected, on four crossings between 216 and 220 GHz in two of this object's eight blocks: ordinary control rings, drifts far from zero, and a visibility fit placing two of the four at the star and two not. A transition of the 225-species list lies within 15 km s^{-1} of every row of Table 4, which 24 976 transitions guarantee and which identifies nothing. We have no explanation, and these are the two objects a repeat observation should point at – HD 14055 for drifts that are lower bounds, ALMA J153702653−33192492 for a radial velocity and one deeper epoch at 216–220 GHz.

4.8. *A stellar-frame multi-epoch stacked search*

Combining the epochs of a star in that star's own rest frame deepens the limit. 56 of the 83 surveyed systems hold more than one searched execution block. 59 of the

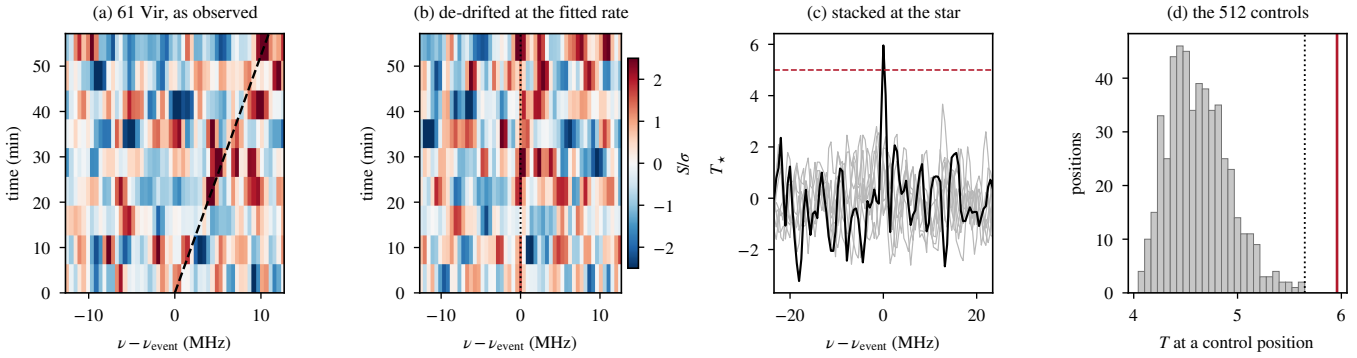


FIG. 6.— **The strongest unattributed crossing of the quality-passing data, and the control field it leads.** Frequencies are in the star's rest frame. *(a)* The dynamic spectrum at 61 Vir's position around the crossing at 346.3232 GHz, observed 2017 May 20, the fitted drift of $+3182\,\mathrm{Hz\,s^{-1}}$ dashed, 225 integrations binned into 9 and plotted in units of each bin's noise. *(b)* The same rows shifted by that drift. *(c)* The de-drifted stack at the star (black) and at the 8 control positions with retained dynamic spectra (grey), in units of its own noise, the trigger dashed. *(d)* The same statistic at each of the 512 rank-screen control positions: the star (solid) against the highest control (dotted).

TABLE 5
CAN TWO CROSSINGS BE ONE CARRIER? THE 13 PAIRS TOWARD ONE STAR WITH AN UNATTRIBUTED MEMBER (§4.6); THE NOTE BELOW DEFINES THE COLUMNS AND THE VERDICTS.

Star	blocks	Δt (d)	$\dot\nu_{1\star}$ ($\mathrm{Hz\,s^{-1}}$)	$\dot\nu_{2\star}$	$\Delta\nu$ (MHz)	miss	one grid	carrier?
61 Vir	X822/X82f	3.00	+1092	+3221	+755.4	15	open	
J1537−3319	X10359/X398	123	−2000	+2390	+223.5	3.6	open	
AU Mic	X1838/X319	310	+1440	+2645	−159.0	–	open	
HD 14055	X2b2ee/X2c3bb	0.09	−2390	+4216	+825.2	2447	no	
	X24be0/Xd46	1.92	+1134	−3823	+63.0	41	open	
	X2c3bb/X3835	2.00	+4216	+1800	−514.6	144	edge	
	X2b2ee/X3835	2.09	−2390	+1800	+310.6	48	open	
	X3835/Xd526	607	+1800	−2563	−555.5	–	open	
	X2b2ee/Xd526	609	−2390	−2563	−244.9	–	open	
	X3835/X25c5e	649	+1800	+2611	+423.3	–	open	
	X2c3bb/X25c5e	651	+4216	+2611	−91.3	–	edge	
	X2b2ee/X25c5e	651	−2390	+2611	+733.8	–	open	
β Pic	X3a90/Xa9df	1.02	−1611	−679	+85.5	132	open	
HD 285968	X108f/X5b4e	4.99	−29	−29	−0.002	0.6	yes	

Starred rates are in the star's frame, the observatory's own acceleration removed (observatory frame: Appendix D.6). *Miss* measures departure from constant acceleration, in units of the drift grid's resolution over Δt; a dash marks a pair that resolution cannot separate. The verdict is set by the bounded-trajectory test, which a constant-acceleration failure does not decide – a miss of 3 or less does, and sets *yes*; *edge*, one rate lies on the grid's outermost trial and is a lower bound. Below the rule, a control: the CO(2–1) line of HD 285968, which does not drift.

TABLE 6
THE TWO EVENTS, THE CROSSINGS WHOSE STELLAR STATISTIC EXCEEDS ALL 512 CONTROL POSITIONS OF THEIR OWN WINDOW. §4.5 READS THEM.

	61 Vir	CP−72 2713
Frequency, stellar frame (GHz)	346.323236	344.296455
Drift rate ($\mathrm{Hz\,s^{-1}}$)	+3182	−2056
$T_\star$	5.958	5.809
Highest control T	5.65	5.68
Fitted amplitude (mJy)	118.8 ± 19.9	11.3 ± 2.0
Peak-channel EIRP (W)	5.1×10^{14}	8.9×10^{14}
Repeat epochs measured	3	5
Nearest covering epoch, separation (h)	−1.97	+2.05
T expected if persistent, $T_{\rm pers}$	9.70	6.30
T measured, matched drift, $T_{\rm rep}$	+0.39	−0.82
Amplitude still bounded, $(T_{\rm rep}+5)/T_{\rm pers}$	0.56 S	0.66 S
Calculated recovery chance Φ at $5S/T_\star$ (at S)	1.00 (1.00)	0.66 (0.90)
Disposition	not confirmed	not confirmed

surveyed stars can be stacked, 5 of them from blocks beyond the census. Over 320 stacked groups, 2219 epoch-windows and 1431 h the flux-density limit improves on each star's best single epoch by a median factor 1.52 and by 4.45 at most. A further 10 stars, all control stars beyond 40 pc, are stacked and excluded from those figures. *A fixed stellar-frame frequency is not a fixed observed frequency, so the epochs are registered before they are added.* The barycentric term alone swings by a median $8.2\,\mathrm{km\,s^{-1}}$ between the epochs of one star, many channels at the 488 kHz Class A width. Each epoch is therefore transformed to the stellar rest frame, divided by its own primary-beam response and combined by inverse variance over the channels of common stellar-frame coverage, keyed on the star's catalogued identity. The transform is verified on a real line, HD 285968's, which 4 epochs in 44 d observe $12.2\,\mathrm{km\,s^{-1}}$ – 19.26 channels – apart and all place at one stellar-frame frequency to $0.03\,\mathrm{km\,s^{-1}}$. Every stacked limit is measured by injection into the stack itself and never read off a stacked spectrum as a threshold. The criterion is evaluated identically at the stacked control positions, so that the whole test's false-alarm rate is measured. *The gain is* $\sqrt{N_{\rm eff}}$ *and not* $\sqrt{N}$, where $N_{\rm eff}=\sigma_{\rm min}^2\sum_e\sigma_e^{-2}$ is the equivalent number of equal-depth epochs. The median $N_{\rm eff}/N$ is 0.65 and the realised gain against $\sqrt{N}$ is 0.78. The shortfall is unequal epoch depth, so an epoch count alone misleads.

The deepest stacked limit is Proxima Centauri's, and it is a Class B limit. Its 25 census blocks are all coarse 15.625 MHz windows. The stack therefore constrains an unresolved spectral excess, discriminates no drift and reaches $\mathrm{EIRP}_{90}=2.2\times10^{13}$ W. *Every stacked limit holds only for an emitter steady in the stellar frame to within one of its own channels, and Proxima Centauri b does not satisfy that.* At 11.185 d about a star of $0.122\,\mathrm{M}_\odot$ the planet moves at $47\,\mathrm{km\,s^{-1}}$. These are the archive's default parameter set (Akeson et al. 2013), for the planet of Anglada-Escudé et al. (2016) and a host mass agreeing with Kervella et al. (2017). A transmitter on the planet therefore swings over $94\sin i\,\mathrm{km\,s^{-1}}$, some 5 channels at $\sin i=1$. The stack cancels such a carrier instead of building it unless the orbit is seen nearly face-on ($\sin i\le0.22$, 2.5 per cent of randomly oriented orbits). Within a single block the same transmitter gains at most

$1.2\,\mathrm{km\,s^{-1}}$, 6 per cent of one channel. For that reason the single-epoch limit of 5.51×10^{13} W – the deepest single epoch in the archive, from a reserved block and not a census limit – is the one that applies to it, and the deepest limit this survey places on the nearest plausible transmitter site. The same tolerance divides the sample. A fine-channel stack admits a far smaller stellar-frame excursion than a coarse one, so the coarse stacks buy reflex tolerance with bandwidth. No crossing of §4.6 drifts slowly enough to survive a fine-channel stack, and the single-epoch limits of §5.3 are the ones that apply generally. *2 groups pass the full criterion and both are known lines*: β Pictoris carbon monoxide in two transitions. That is the astrophysical positive control of §4.4, passing in the stack as well. The survey's own $\pm 50\,\mathrm{km\,s^{-1}}$ mask leaves none of them unattributed.

5. DISCUSSION

5.1. *What limits this search*

This experiment's most transferable result is not the non-detection but a measurement of what limits a carrier search at these wavelengths. It is not depth: the power a transmitter must radiate to cross the threshold here is a median 1.8×10^{15} W per system, and more power buys nothing at a frequency the archive never tuned to or never returned to. That inherited coverage sets the limit.

Line emission, and not interference, is the astrophysical false positive here. Catalogued molecular transitions, almost all of them carbon monoxide, account for 18 of the 52 crossings in quality-passing data, and interference for none (Appendix E). Of those attributions 16 carry support beyond the frequency match – a second transition of the same species, catalogued gas on the sightline, or the crossing's own recurrence – while two rest on the coincidence alone. Lines account for the crossings that stand out and chance for most of the 34 that remain (§4.3.1); the remedy is a velocity mask in each star's own rest frame (§4.4). Extended emission is the second discriminant, with no centimetre analogue: a line filling the primary beam raises star and control annulus together, and 10 of the 12 windows that reached the spatial stage are disc or interstellar carbon monoxide (§4.4, Appendix D).

An illustrative engineering benchmark fixes the scale these limits reach. A 12 m dish radiating 1 MW at 230 GHz with an aperture efficiency of 0.7 has a directive gain $G = \eta(\pi D\nu/c)^2 = 5.9 \times 10^8$ and so an EIRP of 5.9×10^{14} W. That is above the single-epoch crossing limit of 16 of the 60 Class A systems that retain unmasked bandwidth. The same gain makes its beam $\Omega = 4\pi/G = 2.1 \times 10^{-8}$ sr, illuminating 1.7×10^{-9} of the sky with a half-power width of 25 arcsec. These limits therefore apply to beamed transmitters whose beam contains the Solar System, and EIRP is not transmitter power.

The channelisation is inherited. The carrier search rests on the 405 fine-channel windows because almost none of the 1258 coarse ones can tell a drifting carrier from a stationary one (§3.1). The widths on offer run from 15.3 kHz to 31.25 MHz, 3.3 orders of magnitude, or 7 beside the 2.8 Hz channels of centimetre-band carrier searches (Fig. 1(b)), and nothing in the search could choose them.

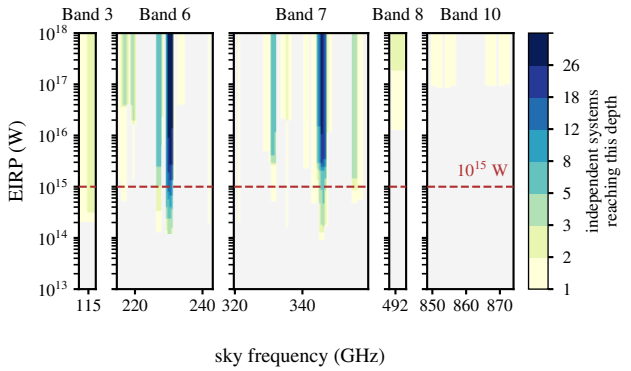


FIG. 7.— Where the experiment could have found something. Shading, the independent systems recovering an unresolved carrier of the plotted EIRP and sky frequency nine times in ten *in a single epoch*; unshaded, none. Confirmation needs a second epoch covering the same frequency, which no power supplies. Panels as in Fig. 2b, which gives that coverage. Dashed, 1×10^{15} W.

Repeat epochs turn a crossing into a testable claim, and an archive supplies them only incidentally. Let the *confirmation coverage* $C_{\mathrm{conf}}(\nu)$ be the number of systems toward which two execution blocks of different observing sessions more than a day apart both cover the sky frequency ν. A carrier found once there could then have been sought again. It must be stated against frequency and not as a fraction of the sample, because repeat visits re-tune and a second epoch is of no use at a frequency it does not cover. Its support is 24.0 GHz of union bandwidth, 50 per cent of the 47.7 GHz searched, falling to 8.5 GHz beyond a year. At its best, near 231 GHz, it reaches 27 systems (Fig. 2b). A sensitivity limit and a confirmation limit are therefore different statements: of the 61 systems, 40 have a second session covering a frequency of the first more than a day away and 13 more than a year away, while for the remaining 21 no transmitted power could have made a confirmation possible.

5.2. *The search space this archive opens*

The one other millimetre search of more than a single line, Mason et al. (2025), covered 117 MHz of Band 3 toward far more distant stars (§1). Band 3 is searched here, in 112 windows down to 90 GHz including β Pictoris' CO $J = 1-0$ control.

Union bandwidth overstates what that buys, because different frequencies were searched toward different numbers of stars and to very different limits. Figure 7 gives instead the number of independent systems that would have yielded a recovery at ninety per cent completeness, against sky frequency and transmitter power. At an EIRP of 1×10^{15} W the covered frequency falls to 19.0 GHz, 40 per cent of the union. That coverage reaches 24 systems, and 16 at the best-covered frequency near 230 GHz. With no limit on power at all, that frequency reaches 38 of the 61 systems, and half the sample is reached only above 3.9×10^{15} W.

On the continuous-waveform transmitter figure of merit of Enriquez et al. (2017), with the single survey-wide fractional bandwidth replaced by the star-bandwidth product an archive requires, this experiment is 4.0×10^4 on the median system and 9.5×10^5 on

the sample's largest threshold power.[5] That is a factor 2.4×10^5 above the Enriquez value once both are read at the same threshold. It splits into 2.76 decades of haystack and 2.61 decades of depth: a far smaller haystack than a wideband centimetre survey of thousands of stars, and a shallower one.

5.3. *What this experiment excludes, and what it does not*

What is not constrained. Every limit here applies to one kind of signal: an unresolved carrier at the stellar position, confined to a single 15.3–1953 kHz channel, lying outside the molecular-line mask, radiating while one of the archival blocks is on sky, and drifting linearly at no more than its window's ceiling, 12.0–13.3 nHz. The grid contains every known planet of the sample except TRAPPIST–1 b, whose windows carry a widened grid reaching it only to the precision its host's mass and orbit are known (§3.1), but a transmitter on a rapidly rotating planet spends most of its reach on spin alone (Fig. 3). Emission broader than a channel needs close to 32× the power for a feature as wide as the widest line in these fields, and a carrier on for part of a block comes back at that fraction of its amplitude. The result is a non-detection and not an exclusion of persistent transmitters: where the archive observed a frequency once no power makes a confirmation possible; a shallow repeat tests nothing, as for 8 of the 33 crossings read at a predicted cell; and the single-cell reading tests only a carrier returning to the same stellar-frame frequency, which an intermittent, accelerating or deliberately re-tuned transmitter need not do (§4.6). Nothing is constrained inside the $\pm 50\,\mathrm{km\,s^{-1}}$ mask, or toward the stars within 40 pc that ALMA has not observed, almost all of the 17 566 this volume holds.

Two blind spots. The analysis forms (XX+YY)/2, which retains the full intensity of a circularly polarised carrier. The loss is not sensitivity but the discrimination polarisation would offer against unpolarised astrophysical emission. ALMA's feeds are linear, and the circularity lies in the cross-hands, which this analysis does not extract. The frequency coverage, too, is the archive's and not a choice. The 203 GHz positronium hyperfine line that Steffes & DeBoer (1994) and Mauersberger et al. (1996) observed lies in ALMA Band 5. This survey holds 4 coarse-channel windows there, all toward α CMa B, and none of them covers it.

5.4. *What a dedicated ALMA technosignature experiment should do differently*

The changes a purpose-built experiment on the same telescope needs follow from the limitations above.

- *Channelise finely everywhere.* The 1258 coarse-channel windows cannot carry a carrier search at all.
- *Choose the targets and the frequencies.* An archive weighted toward dusty discs is the worst case for molecular-line confusion. The M dwarfs that dominate the nearby habitable-zone planet census are the least covered. Windows placed deliberately away from the bright transitions would recover most of the 2.38 GHz of the 47.7 GHz Class A union that the mask costs.
- *Repeat the epochs on purpose, at the same tuning.* Two or three visits separated by days and by years turn a crossing into a testable claim, and the separation is a scheduling constraint rather than extra sensitivity (Fig. 2b). Two epochs cannot test an orbit, only fit one. Three separated epochs over-determine it, and 27 of the 61 systems here already carry three at a common frequency. The shortfall is therefore not coverage but a candidate seen at more than one of them.
- *Retain the visibilities.* Rotating the phase centre to the star leaves a genuine point source entirely in the real part of the visibility. A test on that part asks whether the data contain a source where the star is, a stronger question than any rank against neighbouring positions in an image.

6. CONCLUSIONS

1. *What is new.* We searched 405 fine-channel ALMA windows toward 61 systems within 40 pc for a drifting narrowband carrier at 114–873 GHz: 47.7 GHz of unique frequency in 26 separate intervals, at least 1363 times either earlier millimetre search of nearby stars (Steffes & DeBoer 1994; Mauersberger et al. 1996), with 391 windows long enough for the drift to cross a channel (§2).
2. *How deep.* The median system's best window would have registered a carrier of 1.8×10^{15} W nine times in ten, and 24 of the 60 systems with a constrained window reach 10^{15} W, or 22 allowing for atmospheric decorrelation; a 12 m dish radiating 1 MW at 230 GHz, beamed at the Sun, lies above the limit for 16 of them. For 54 of the 60 systems they are transferred from injections into other windows (§4.1). For Proxima Centauri the census stack reaches 2.2×10^{13} W for an emitter steady in the star's own frame; for a transmitter on Proxima Centauri b the archive's deepest single epoch, 5.51×10^{13} W and not a census limit, applies (§4.8).
3. *What was found.* Of the 52 quality-passing crossings, 18 are attributed to catalogued molecular lines, mostly carbon monoxide, 16 with independent support (§5.1). Of the other 34, one is atmospheric and two lie in the archival tail; the remaining 31 compare with the 24.8 predicted from control positions beside the star – within what chance allows, and leaving room for a further 17 (§4.3.1).
4. *What the repeat observations show.* No crossing satisfies the confirmation criterion, and the criterion is narrow: it reads the one cell a discovery predicts, so it recovers only a carrier that returns to within half a native channel – as a spectral line does, and as the 33 tested, accelerating at 0.48 to $3.64\,\mathrm{m\,s^{-2}}$, do not

[5] The denominator is the star-bandwidth product $\sum_i \Delta\nu_i/\nu_i$, one term per system, in which $\Delta\nu_i$ is the union of that system's own searched windows and ν_i the midpoint of the span they cover. It is 0.64 over the 60 systems against 5.8 on a single survey-wide fractional bandwidth. The two are a factor 9.1 apart because the median system was covered over only 1.84 GHz of the 47.7 GHz union. An archive searches different frequencies toward different stars, which a single fractional bandwidth credits to every system alike. The published values are read at the common 5σ threshold of Fig. 1, where 0.85 becomes 0.17 and 0.14 becomes 0.072. Margot et al. (2023) publish no figure of merit, and the value quoted here is the same expression on the 11 680 stars, 1.15–1.73 GHz and EIRP $> 1.35 \times 10^{13}$ W they do give.

unless a periodic motion is caught at the matching phase. Where it applies it works: 13 of the 15 line-attributed crossings return. Searched at every drift across each covering window, 17 of the 212 windows reach the trigger against 11.3 expected ($P = 0.05$; 10.4 at $P = 0.03$ from control positions), a marginal excess with 10 of it toward two objects; the grid separates 1 of 13 crossing pairs and leaves the rest open. No transmitter is excluded by any of this. Confirmation is limited by coverage – 21 systems have no repeat at a covered frequency – and by the criterion, not by depth (§5.1, §4.6).

5. *What is not constrained, and what to do next.* The limits apply only to a narrowband carrier outside the molecular-line mask, on while ALMA looked and drifting by at most 12.0 nHz, 6 per cent of the recommended ceiling (Sheikh et al. 2019), and only to stars ALMA observed (§5.3). Other observers' choices of channels and repeats, not collecting area, limited the search. A dedicated programme would channelise finely, avoid the molecular lines and revisit each target at one tuning days and years apart, beginning with the two objects that carry the excess: HD 14055, the survey's most-observed star, and ALMA J153702653−33192492, whose share is unexplained but not significant once its selection is allowed for (§4.7, §5.4).

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This paper makes use of the following ALMA data: the 67 projects listed in Table A3, each cited as ADS/JAO.ALMA# followed by its project code. The 407 primary-census execution blocks searched span 2013 October 6 to 2025 July 11, and we thank the principal investigators and teams who proposed and obtained them. ALMA is a partnership of ESO (representing its member states), NSF (USA) and NINS (Japan), together with NRC (Canada), NSTC and ASIAA (Taiwan), and KASI (Republic of Korea), in cooperation with the Republic of Chile. The Joint ALMA Observatory is operated by ESO, AUI/NRAO and NAOJ. This work has made use of data from the European Space Agency (ESA) mission *Gaia* (Gaia Collaboration et al. 2023), processed by the *Gaia* Data Processing and Analysis Consortium (DPAC). This research has made use of the SIMBAD database (Wenger et al. 2000) (CDS, Strasbourg, France); of the NASA Exoplanet Archive (Akeson et al. 2013), which is operated by the California Institute of Technology under contract with the National Aeronautics and Space Administration under the Exoplanet Exploration Program; and of the Splatalogue spectral-line database (Remijan et al. 2007).

Use of generative artificial intelligence. Large language models were used in this work: they wrote most of the software – the archival retrieval and calibration scripts, the search and vetting pipeline, the statistical tests, and the generators that produce every number, table and figure in this paper. They also drafted text, which the authors revised. The authors set the scientific question, chose every criterion, and are responsible for the content of this paper.

Testing and validation of that analysis. No result here is accepted on a model's assertion: every number in the text is written by the program that computes it from the data rather than transcribed, which removes transcription error and nothing else, and a separate audit rederives the identities that must hold among the published values – counts that must partition, quantities that must sum, numbers reached by two routes that must agree – and refuses the paper where one does not. Every assertion in the analysis was first made to fail on purpose, on deliberately wrong input, before it was relied on, since an assertion that has never fired is no evidence that anything is right.

Software. The pipeline depends on CASA 6.7.0 (CASA Team 2022), PYTHON 3 with ASTROPY (Astropy Collaboration et al. 2022), NUMPY (Harris et al. 2020) and SCIPY (Virtanen et al. 2020), and the ALMA Science Archive and SIMBAD TAP interfaces through ASTROQUERY (Ginsburg et al. 2019); all are free and open-source, at the versions cited.

Author contributions. G.J.W. conceived the survey, defined the target sample and the selection criteria, and wrote the manuscript; R.D. built and operated the archival retrieval, calibration and search pipeline, implemented the spatial control statistic and the injection–recovery campaigns, and produced the per-window catalogue and its regeneration scripts. Both authors designed the vetting procedure, chose the masked species, carried out the statistical validation reported in §3.1 and Appendix D, and agreed the dispositions of the crossing ledger.

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Competing interests. The authors declare no competing interests. VBRL Holdings is a privately held company through which R.D. conducts independent research, named here solely as his affiliation; it had no role in the design of the survey, the selection of targets, the analysis, the decision to publish or the content of this paper, and provided no funding for the work.

DATA AVAILABILITY

Public. Every observation this paper rests on is public and is identified here precisely enough to be fetched: the 407 primary-census execution blocks, and those of every other set of Table A1, are held in the ALMA Science Archive under the 67 project codes of Table A3 and are retrieved by the execution-block identifier, which the crossing ledger gives for each of the 56 threshold crossings (Chen et al. 2022); the astrometry is Gaia DR3, the planet counts the NASA Exoplanet Archive, and the mask one Splatalogue query of CDMS and JPL made on 2026-10-06 and repeatable as it stands.

Deposited. The machine-readable data behind this paper are deposited as a citable archive (White & Dey 2026): the per-window catalogue of all 1663 searched windows; the Class A and Class B sensitivity measurements; the 56 threshold crossings with their molecular-line coincidences, the atmospheric flag and both field readings of the trigger; the repeat-observation readings; the injection trials and the completeness calibration carried from them; the control-field statistics, including the above-threshold cell counts below the trigger; the execution-block and count accounting, all 30 rows of Table A2 included; and the data behind the figures – every

table describing its own columns and units and carrying the project code and execution block that tie each row to an observation, five of them also accompanying the paper on the arXiv as ancillary files. The archive also holds the software. The search pipeline is deposited as it ran – the archival retrieval, the calibration replay, the direct-Fourier extraction, the de-drifting search and its control ring, the injection–recovery campaign and the visibility-domain fit – together with the generators that produce the crossing ledger, every table and every number printed here, the configuration they read, the versions of CASA, PYTHON, NUMPY, SCIPY and ASTROPY they ran under, and instructions for taking one public ALMA execution block from the archive to a statistic printed in this paper. Short reference implementations of the four definitions the results rest on – the statistic of Equation 1, the control-position generator, the injection–recovery harness and the line mask – are deposited beside the pipeline and are runnable against the deposited tables on their own. The deposit records the defects of that software that we know of rather than removing them, since code that has been tidied is not the code that produced the result.

What the deposit supports. With Table A1 for the block and window partition and Appendix F for every system of the carrier search, it rebuilds the window, star and system counts, the crossing counts and their classification, the chance expectation of 24.8 crossings as a sum over each Class A window's own control exceedance rate, masked share and atmospheric factor, the EIRP_{90} summaries including which windows carry an injected completeness and which a transferred one, and every exclusion depth with its recovery probability. Two quantities it does not give on their own: the interval on the chance expectation, whose resampling draws on quantities that are not per-window, and the completeness of the individually injected windows, whose injection was made into per-integration spectra that are not deposited.

Calibration. ALMA's flags and the legacy blocks' flux scaling were not applied; the measured effect is in §3, and a recalibration with both applied is the remedy. *Not public.* The calibrated visibilities and the per-window products are not distributed, so this execution of the search cannot be repeated: the visibilities of the 407 blocks are of order a terabyte, and the extracted per-window spectra were discarded as the search consumed them, which was a constraint of the volume available and not a choice about access. Nothing is withheld, and every observation is identified precisely enough to be fetched and recalibrated by the deposited code. What the deposit does support is narrower and is worth stating exactly: the published statistical summaries rebuild from the deposited tables row by row; the generators reproduce every number, table and figure here from the deposited products; and one public execution block runs from the archive through the deposited pipeline to a statistic printed here. Reproducing this search over the whole archive would mean re-fetching and recalibrating it, which is a new execution and not a repetition of this one.

APPENDIX

A. SPECTRAL RESPONSE, FREQUENCY CONVENTIONS AND THE PRIMARY-BEAM RESPONSE

A.1. *Frequency frames, linewidth and smearing*

Every frequency axis here is *topocentric*, as the correlator delivers it; velocities enter only at vetting, where the line mask carries each crossing to the stellar frame (Appendix C). The native channel width is the linewidth convention for every threshold: $\mathrm{EIRP}_{\mathrm{trig}}$ is the minimum *total* power of a channel-confined carrier, never a spectral power density, and spreading it over N channels lowers the statistic as $N^{0.81}$, so a feature as wide as the widest line these fields resolve (69 channels) must carry $32\times$ a carrier's power. The Hz-resolution surveys' limits are total powers in the same sense, which makes the axes of Fig. 1(a) commensurable.

ALMA applies online Hanning smoothing by default (Cortes et al. 2026), so neighbouring channels are correlated: for the 0.25/0.5/0.25 kernel $\rho_1 = 2/3 = 0.667$, $\rho_2 = 0.167$ and $\sigma_{\mathrm{smoothed}}/\sigma_{\mathrm{raw}} = 0.612$, and the peak channel holds 0.50 of an unresolved carrier's flux at a channel centre and 0.42 on a boundary, a median 0.48 over sub-channel phase. *That loss enters the limits exactly once; the thresholds are free of it and the limits are not.* A window's trigger flux is the bare 5σ of its own spectrum, so the *total* power of a channel-confined carrier set against it would be optimistic by $\times 2.00$–$\times 2.36$. The injected carriers are not set against it in that way: each is deposited with the correlator's own channel response, at a sub-channel phase drawn afresh, and recovered through the same trigger, which reads the peak channel. The measured completeness therefore already contains the loss, and factorises into it exactly: $3.03\,P_{\mathrm{trig}}$ is $\times 2.29$ for the channel response and $\times 1.325$ for the margin a peak channel needs above the nominal trigger for nine carriers in ten to return. So P_{trig}, the sensitivity S_{min} below and the filled symbols of Fig. 1(a) are nominal throughout, which is what makes them commensurable with other programmes' detection thresholds, while EIRP_{90} carries the correction once. That factor is 1/0.4375, the phase-mean peak-channel share a deposited tone keeps, 8 per cent short of the exact response's 0.4741, so the completeness is conservative by that much.

Intra-integration smearing, which the injections do not model, is divided out of every limit once as η_{smear}, the mean channel response over the sweep within one integration; it exceeds one per cent in 9, reaching $\times 1.05$, and is measured for 1054 of the 1663 windows and bounded for the other 609 at 11.6 s, the longest integration delivered. Re-scoring the 49 injected windows with every tone de-rated by its own η_{smear} gives a penalty never larger than the one applied, so the correction errs the safe way; the full treatment is in `documentation/appendix_A_smearing.md` of the data archive (White & Dey 2026).

A.2. *The primary-beam response*

The sensitivity quoted for a window is $S_{\mathrm{min}} = 5\sigma/A$, where A is the primary-beam response at the star's offset

from the phase centre. Archival fields are pointed at catalogue positions, so these high-proper-motion stars can sit well off axis – a median 0.089″ and at most 17.99″ over the 1663 windows, A falling to 0.2308 and below 0.9 on 46 – each a J2000 pointing on a fast-moving star rather than an error.

The response applied is a Gaussian of the full width the observatory documents, $1.13\,\lambda/D$ (Cortes et al. 2026), at each window's own dish diameter. It matches the blocked-aperture response CASA applies (CASA Team 2022) to 2 per cent in width; the two part company with offset, which sets the retention rule of §2, and *the furthest census window lies at* $0.82\,\lambda/D$*, where they agree to 6.2 per cent, the largest beam-model uncertainty any sensitivity here carries.*

The correction is applied once: $S_{\min} = 5\sigma/A$ closes on all 1663 catalogued rows to 8.7×10^{-5}, and to 8.7×10^{-5} on the 1653 whose A is not back-solved from $S_{\min}$, while the statistic, ranks and attribution are ratios within one window, in which A divides out.

The annual parallax is measured rather than modelled: a carrier injected at the full apparent position of the 8 furthest-out windows and recovered from the same visibilities returns the normalised dirty beam, 0.92 against a predicted 0.89 at 0.25 of a beam and 0.58 against 0.49 at 0.67, so a limit corrected by the model is shallower than it need be. ϵ Eridani, with 24 of its 44 windows 0.75–$0.82\,\lambda/D$ off axis and the largest parallax in the search, has no parallax record, so its limit is optimistic by at most 9 per cent.

A component of a bound pair does not travel in a straight line, and the propagation assumes it does. 30 of the 91 searched positions have a companion within 400 arcsec (Mason et al. 2001; Matson et al. 2026), the displacement taken from a visual orbit or bounded from the separation, with masses from Pecaut & Mamajek (2013), or from the proper-motion anomaly (Kervella et al. 2022); the largest is 0.20 arcsec at UCAC2 20312880, 0.28 of a beam, and no system's limit moves by more than 16 per cent. The exception is Barta 161 12, HD 155555C and LHS 1140, whose solutions are disturbed with no companion and no Hipparcos epoch, so the 24 windows of those 3 stars carry limits optimistic by an unmeasured amount.

Table A1 accounts for every execution block the query exposed and Table A2 for every event; their itemisation, all 30 rows, accompanies this paper as data under the same row numbers. Three counts in the second table are 52 over three different sets – the adopted ledger's Class A crossings, the per-window catalogue's and the quality-passing ones – and the first two differ on 6 rows, where a narrower-field reading falls on the other side of the trigger (Appendix D). The census's on-source time is 281 h; planet counts are from the NASA Exoplanet Archive `pscomppars` table (Akeson et al. 2013), accessed 2026 October 4.

B. THE SPATIAL RANK, AND WHAT IT CAN SUPPORT

The spatial rank asks whether an excess sits at the star or anywhere else in the field. It is a diagnostic on which no disposition rests (§3.2), and this appendix is the measurement that settles why.

The star lies inside the control annulus at the phase centre, and the displacement is real: the star carries more above-threshold cells than a control position at every level below the trigger (Appendix D). *It is not a primary-beam effect*: spectra are extracted in apparent flux density from phase-rotated visibilities (Appendix A), and a noise rise toward the field edge would put the stellar rank *above* one half, the opposite of what is measured; the ensemble's own weak rise with radius is removed by standardising each probe within its radius bin. The full treatment is in `documentation/appendix_B_spatial_normalisation.md` of the data archive (White & Dey 2026).

A calibration error that varied with frequency would explain the displacement, and it is the only mechanism on offer. A bandpass residual, or a coarse-mode system temperature interpolated near a telluric line (Hunter et al. 2018), survives the per-integration median along frequency, multiplies the continuum, so acts at the star and not on empty sky, and stands still in topocentric frequency, stacking at near-zero drift and putting the stellar rank *below* one half, the sign observed. *Amplitude excludes it.* A fractional error ϵ needs $\epsilon = 5\sigma/S_{\rm c}$ to reach the trigger, with the continuum $S_{\rm c}$ measured for 1557 of the 1663 windows and checked against independent continuum images. No window requires as little as the 5–20 per cent a coarse system temperature leaves at an ozone line – the least anywhere is 21.4 per cent, the median 1067, and 53 for the least demanding of the 51 crossings with a continuum – the bandpass branch needs 21× the 1 per cent quoted for the worst bands (ALMA Science Portal 2026; Kameno 2014), and 1 crossing lies within 62.5 MHz of an atmospheric transition (Appendix E). The stellar statistic over the median control statistic has median 1.001 (0.994–1.005) over 1609 census windows without an adopted crossing, so the two noise scales agree.

B.1. *The reserved hold-out, and what it calibrates*

The statistic and the mask are set on the searched sample and cannot calibrate themselves, so a further sample is held out by an identifier rule, $\mathrm{sha256}(\mathrm{UID})_{[:8]} \bmod 5 = 0$, with two guards using identifiers alone: a block on a named list stays in the survey, and no system loses every block. Its independence of the statistic and the mask is our assertion, not something a reader can check. 77 of 484 blocks are reserved, 315 windows toward 36 stars, with no star or system lost. Because the rule hashes the block and not the session, 64 per cent of the reserved set shares calibrators, weather and setup with a searched block – a session-level hash would have been stronger – so every quantity from it is also given over the 28 blocks that share no session; on those the control level has a radial slope of -0.043 ± 0.028.

B.2. *The recurrence statistic under the null*

The bound in the crossing ledger is one formula. Writing S and $\sigma_{\rm d}$ for the discovery amplitude and its error, and $\bar{a}$ and $\bar{\sigma} = (\sum_i \sigma_i^{-2})^{-1/2}$ for the inverse-variance combination of the repeat readings at the predicted channel,

$$\sigma_{\rm excl} = \frac{S - \bar{a}}{\sqrt{\sigma_{\rm d}^2 + \bar{\sigma}^2}} = \frac{T_{\rm pers} - T_{\rm rep}}{\sqrt{1 + (T_{\rm pers}/T_\star)^2}}, \qquad \text{(B1)}$$

TABLE A1

The survey ledger, part one: execution blocks and windows. Each row carries its own number, and a group sums to the row whose number heads it. Hours are window-hours, so the classes partition the hours as they partition the windows. Rows 3.2 to 3.5 are the archival gap, over which no result is formed.

	Set of execution blocks	Blocks	Windows	Stars	Systems	Hours
1	Archival scope: blocks the query exposes	656	–	–	–	–
1.1	… processed with the same pipeline	479	–	–	–	–
1.2	… not processed at the freeze	177	–	–	–	–
2	Processed with the same pipeline[c]	484	–	–	–	–
2.1	**Primary census**	407	1663	90	83	1098
2.1.1	Class A, fine channels[a]	–	405	66	61	228
2.1.2	Class B, coarse channels[a]	–	1258	89	82	870
2.2	Reserved hold-out	77	315	36	–	233
3	Not processed at the freeze: what became of each	177	–	–	–	–
3.1	Calibrated and searched	144	–	–	–	–
3.1.1	**Archival extension**, reported	115	463	16	16	360
3.1.1.1	Class A, fine channels[a]	–	67	11	–	48
3.1.2	Searched after the extension was frozen	8	32	–	–	–
3.1.3	ϵ Eri, both pointings[d]	8	32	–	–	–
3.1.4	No window reached the export	13	0	–	–	–
3.2	No calibration in the ALMA delivery	18	–	–	–	–
3.3	Calibration failed	11	–	–	–	–
3.4	Not attempted	3	–	–	–	–
3.5	Working disk insufficient	1	–	–	–	–
4	Beyond 40 pc: out-of-sample control	47	198	22	–	88
5	Rank-calibration sample[b]	326	1322	56	–	–

[a] The two classes partition the census windows, not its blocks: 255 blocks carry a Class A window and 392 a Class B one, and most carry both, so the hours of the two rows partition the windows and not the observing time.

[b] Not an out-of-sample set and not an addition to this ledger: every one of its blocks is counted in a row above, in the primary census or in the hold-out, and it is listed apart only to keep it from being counted twice.

[c] Panels 1 and 2 overlap rather than add: 479 of these 484 blocks are row 1.1, and the other 5 carry no member observing unit in the archive metadata, so the scope query does not expose them; 4 of the 5 are in the primary census.

[d] Every one is an ϵ Eri Band 6 block. The archival pointings put the star either at the centre of the field, at a response of 1.00, or well off it, where it is 0.23–0.28 over the 16 of this row's 32 windows that lie off axis, above the 0.22 retention floor of §2 in every case. All of them are searched, each from its own recorded phase centre, and their conventions are in Appendix A.
The same band's 12 windows that fall in the primary census are counted in row 2.1 and carry every census statistic; the on-axis pointings reach a response of 1.00.

with $T_{\rm pers} = S/\bar{\sigma}$ and $T_{\rm rep} = \bar{a}/\bar{\sigma}$, so every bound follows from Eq. B1 and three columns of that table. $T_{\rm rep}$ is one cell at the carried drift; the maximum over every drift at that cell, tabulated beside it, is never compared with $T_{\rm pers}$, being positive on average whatever is there, and as an amplitude selected from 10^2 to 10^7 cells at $T_\star \simeq 5$ is biased high, the limit is stated as $(T_{\rm rep}+5)/T_{\rm pers}$. The reading is centred on zero, scattering by 0.999 about +0.002 over 484 single-cell readings of every repeat block in the census and hold-out, less the 4 of the crossing that recurs, a molecular line. The bound's power is $\Phi(T_{\rm pers}-5)$ at the discovery amplitude and $\Phi(5T_{\rm pers}/T_\star - 5)$ at the trigger amplitude $5S/T_\star$, the faintest that could have produced the crossing, and the 8 where these are smallest are reported as untested; each crossing's measured recovery chance accompanies this paper as data.

C. THE LINE MASK

A crossing that falls on a known molecular transition has an astrophysical explanation to hand, and the mask says which crossings those are.

The list. The mask holds every transition of the 11 species these setups cover – the three CO isotopologues, HCN, HCO^+, CS, CN, SiO, H_2CO, [C I] and hydrogen recombination – inside the 46 searched intervals below 150 K in upper-state energy, 96 transitions (Table C1), main hyperfine components only and no strength cut, from one Splatalogue query (Remijan et al. 2007) of CDMS and JPL (Pickett et al. 1998; Müller et al. 2005) dated 2026 October 6.

The species are the ones the observers went to the telescope for, and that is a limitation: 38 per cent of the 405 Class A windows are centred within $\pm 50\,{\rm km\,s^{-1}}$ of a listed transition, against 1–8 per cent for the same windows displaced, so *unattributed* means unattributed against the observers' own list. A wider list – every transition below 150 K of all 225 species with an interstellar or circumstellar detection, 24 976 in all – covers 98 per cent of the Class A union against 5.0 per cent for this one and attributes all 38 unattributed crossings with 100 per cent of the control exceedances, emptying the category for star and control alike; at that density proximity is not an identification.

The pair that comes closest to an identification. The 2 unattributed crossings of β Pictoris lie 8.3 and $8.8\,{\rm km\,s^{-1}}$ in the star's frame from methanol $5_K \rightarrow 4_K$ components, offsets agreeing to $0.48\,{\rm km\,s^{-1}}$ (0.7 per cent by chance). Three spectroscopic measurements decide against them: the brightest member, $5_0 \rightarrow 4_0$ A$^+$, would have been the window maximum and is not; the weaker component is at 0.94 of the stronger where no temperature allows more than 0.65; and both drifts lie 71 and 30 grid steps from the zero a line at rest requires. Whether the two could be one carrier is argued in Appendix D.5. The full treatment is in `documentation/appendix_C_line_mask.md` of the data archive (White & Dey 2026).

Attribution is decided in the star's own rest frame, and the sign is not the usual one. A crossing's frequency is carried from the topocentric value to the barycentre using its own block's pointing and mid-time, and then to the stellar rest frame using the star's systemic velocity. That frame, and no other, is where a coincidence is judged, a tolerance applied in the observed frame spending most of its budget on the Earth's motion. The 4 HD

TABLE A2
The survey ledger, part two: events, dispositions and confirmation. Every headline count is read from here; rows 2.2–2.4 define each unattributed set.

Count		Value	What it counts
1	Threshold crossings at $T_\star \geq 5$	56	census and archival tail
1.1	Class A, fine channels	52	the carrier experiment
1.2	Class B, coarse channels	4	all in the one block failing §4.3.2
2	Quality-passing crossings	52	the same rows as 1.1
2.1	… attributed to a transition	18	of 18 coincidences inside the mask in the star's own frame; rows 2.1.1 and 2.1.2
2.1.1	… with evidence beyond the coincidence	16	recurrence at the predicted cell, a second transition of the same species, or catalogued gas on the sightline
2.1.2	… on the frequency coincidence alone	2	HIP 10679 and HR 1010: consistent with molecular emission, not identified as it
2.2	… unattributed	34	rows 2.2.1 to 2.2.3
2.2.1	… in the primary census	31	against 24.8 expected by chance at the 405 windows' own control positions – a count chance delivers with probability 0.14
2.2.2	… in archival-tail blocks	2	outside that comparison
2.2.3	… coincident with an atmospheric transition	1	within 62.5 MHz of one in the topocentric frame (Appendix E), and counted as atmospheric on both sides of that comparison
2.2.5	… recovered in a later block	0	no confirmed signal; 13 attributed ones do
2.3	All unattributed crossings, both quality classes	38	rows 2.2 and 1.2 together, every crossing of the failing block being unattributed: the population the recurrence test and the interference screens of Appendix E are formed over
2.3.1	… whose predicted cell was read	37	the remaining 1 retains no discovery dynamic spectrum, so no amplitude could be carried to a repeat epoch for it
2.3.2	… carrying an amplitude bound	33	row 2.3.1 less the 4 of row 1.2, whose discovery amplitude comes from the block that fails the quality criterion
2.3.3	… a catalogued window of a later block covers	37	the set the drift-pair accounting of the data archive is formed over; 1 is covered by no catalogued window at all
2.3.4	… in a block the per-window catalogue describes	36	the image-sideband population of Table E1; the 2 of row 2.2.2 lie outside it
2.3.5	… with both an amplitude bound and a catalogued covering window	32	rows 2.3.2 and 2.3.3 together, the set §4.6 sets against the orbital excursion
2.4	Unattributed crossings of the census and the reserved hold-out in a searched window	40	row 2.3.4 with the 4 of §4.6.1: the population of the sky-frequency occupancy test of Appendix E
3	Class A systems	61	of 83 surveyed
3.1.1	Class A systems with same-frequency coverage in a separated session	**40**	**the only confirmation count quoted**: a later block in another session covering a frequency the first covered, 13 of them over a year later. For an accelerating carrier, confirmation further requires the repeat to fall at a matching orbital phase. Over the census together with the repeat blocks the recurrence test reads
3.3	… with no separated second block at a covered frequency	21	confirmation impossible over the 166 blocks the recurrence test reads; 36 of all 83
4	Class A union bandwidth	47.7 GHz	26 disjoint islands
4.1	… removed by the line mask	5.0 %	2.38 GHz: the figure the text quotes, against 3.1 per cent of the 118.1 GHz archival union

TABLE A3
The 67 ALMA proposals the searched blocks belong to, as ADS/JAO.ALMA#*code*. Every census block belongs to one.

2012.1.00142.S, 2012.1.00198.S, 2012.1.00268.S, 2012.1.00385.S, 2012.1.00437.S, 2012.1.00993.S, 2013.1.00359.S, 2013.1.00588.S, 2013.1.00645.S, 2013.1.01147.S, 2013.1.01214.S, 2015.1.00307.S, 2015.1.00633.S, 2015.1.00783.S, 2015.1.01015.S, 2015.1.01260.S, 2016.1.00104.S, 2016.1.00358.S, 2016.1.00515.S, 2016.1.00637.S, 2016.1.00803.S, 2016.1.00817.S, 2016.1.00878.S, 2016.1.00880.S, 2016.2.00015.S, 2016.2.00200.S, 2016.A.00013.S, 2017.1.00167.S, 2017.1.00200.S, 2017.1.00215.S, 2017.1.00561.S, 2017.1.00698.S, 2017.1.00704.S, 2017.1.00786.S, 2017.1.00986.S, 2017.1.01583.S, 2017.1.01644.S, 2018.1.00045.S, 2018.1.00072.S, 2018.1.00126.S, 2018.1.00461.S, 2018.1.00470.S, 2018.1.00947.S, 2018.1.01149.S, 2018.1.01545.S, 2018.1.01833.S, 2019.1.00189.S, 2019.1.00696.S, 2019.1.01443.T, 2019.1.01517.S, 2019.1.01603.S, 2019.1.01681.S, 2019.2.00208.S, 2021.1.00485.S, 2021.1.01209.S, 2021.2.00097.S, 2022.1.00134.S, 2022.1.00338.L, 2022.1.00793.S, 2022.1.01163.S, 2023.1.00390.S, 2023.1.00487.S, 2024.1.00165.S, 2024.1.00595.S, 2024.1.00722.S, 2024.1.01095.S, 2024.A.00040.S

285968 epochs show the transform working in the only way that can be checked: $12\,\mathrm{km\,s^{-1}}$ apart as observed, they agree to $0.03\,\mathrm{km\,s^{-1}}$ in the star's frame, a twentieth of one of their channels. Three are census crossings and the fourth is a hold-out block. The observatory set all four executions to one source-frame frequency, so the agreement bounds the part of the correction that changes with epoch, which is the large part, and not a constant error in the systemic velocity. Throughout, $\Delta v_\star = c\,(\nu_\star - \nu_0)/\nu_0$, so a positive offset means the crossing lies *above* the transition in frequency, the opposite sign to the velocity convention of a spectral line.

Every crossing is also read in the local standard of rest, where interstellar gas is at rest. Toward ALMA J153702653−33192492, HD 139084 B, HD 285968 and HD 48370 the CO($1 \to 0$) survey of Dame & Thaddeus (2022); Dame et al. (2001) shows emission at the crossing's own LSR velocity (within $3\,\mathrm{km\,s^{-1}}$ of its peak, $\geq$0.23 K), so 6 of the 18 coincidences are interstellar; else-

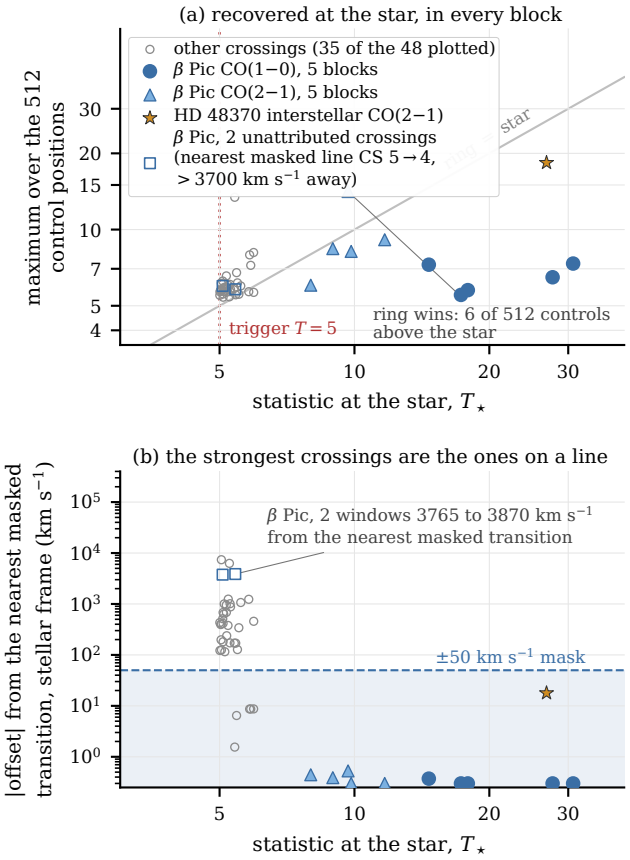


FIG. B1.— The β Pictoris positive control. *(a)* Stellar statistic against the maximum over the window's 512 control positions, filled for blocks recovering its CO. *(b)* Offset from the nearest masked transition in the star's own frame, $\pm 50\,\mathrm{km\,s^{-1}}$ marked: its 10 CO crossings lie 0.1–0.5 $\mathrm{km\,s^{-1}}$ from CO, its 2 unattributed ones 3765–3870 $\mathrm{km\,s^{-1}}$ from any; grey, the other quality-passing crossings.

TABLE C1
THE LINE MASK, BY SPECIES: THE NUMBER N OF TRANSITIONS, THE FREQUENCY RANGE THEY SPAN INSIDE THE SEARCHED INTERVALS AND THEIR UPPER-STATE ENERGIES.

Species	N	ν_0 (GHz)	E_u (K)
H	5	92.03 – 231.90	–
H_2CO	13	101.33 – 491.97	10–106
$C^{18}O$	3	109.78 – 329.33	5–32
CN	50	113.12 – 680.30	5–114
CO	3	115.27 – 345.80	6–33
CS	3	195.95 – 342.88	24–66
SiO	2	217.10 – 347.33	31–75
^{13}CO	8	220.40 – 330.59	16–32
HCN	7	354.50 – 354.51	43–43
HCO^+	1	356.73 – 356.73	43
[C I]	1	492.16 – 492.16	24

where the Planck CO map (Planck Collaboration et al. 2016) stays below $0.178\,\mathrm{K\,km\,s^{-1}}$. HD 48370 is where the two readings part: its coincidence is $-17.8\,\mathrm{km\,s^{-1}}$ from the star's own CO(2 → 1) but at $+23.9\,\mathrm{km\,s^{-1}}$ in the LSR on a cloud line peaking at +21.8, so its extended emission is a cloud and not a disc.

Table C1 gives the mask species by species; the hydrogen recombination lines among them carry no tabulated upper-state energy, and are blank in that column.

The coincidences, and the one species left out. 18 of the 56 crossings fall within the mask in their own star's frame, and these are the 18 attributions. Sulphur monoxide is deliberately not listed – no debris disc has a reported SO detection, so masking its 15 transitions would have declared no constraint over bandwidth the survey does constrain – and the three crossings within the half-width of one (CD−57 1054, CP−72 2713 and HD 161868, at −1.7, −12.3 and $-48.0\,\mathrm{km\,s^{-1}}$) are reported as unattributed, 2 below the rank screen. Nor do the tests made find interstellar SO: in the LSR the offsets are −3.5, −12.8 and $-53.4\,\mathrm{km\,s^{-1}}$, none within the cloud test, and no CO(3 → 2) feature above the trigger accompanies any. The hold-out is treated identically, 2 attributed and 4 unattributed (§4.6.1).

The half-width, and what it costs. Keplerian motion where these molecules survive, $\lesssim 13\,\mathrm{km\,s^{-1}}$ beyond 10 au, sets it at $\pm 50\,\mathrm{km\,s^{-1}}$. *The cost is stated on the union of searched sky frequencies*, the axis on which coverage is quoted throughout: the frequencies within that half-width of a listed transition's rest frequency, merged where transitions overlap, are 2.38 GHz of the 47.7 GHz Class A union, 5.0 per cent, and 3.71 GHz of 118.1 GHz over both classes, 3.1 per cent; the wider list's share is measured the same way. A union over windows toward different stars has no single rest frame, so this describes the coverage and not any one star; a limit is formed instead over its own window's channels outside the mask in that star's own frame (§3.2). Attributing on either frame would mask 21 per cent more for no further attribution.

Of the 2 labelled crossings not recovered (§4.6), one rests on a velocity instead, its sightline carrying interstellar emission $0.5\,\mathrm{km\,s^{-1}}$ from the crossing, with 193–496 of the 512 control positions above the star in every repeat block.

D. THE FALSE-ALARM EXPECTATION AND THE CONTROL ENSEMBLE

D.1. *The expectation, and what it is conditional on*

One attribution set serves both sides of the comparison: the mask of Appendix C, which holds no sulphur monoxide, so the three crossings on SO transitions are unattributed and a control maximum on one is an unattributed exceedance. Each of the 405 Class A windows of the primary census contributes the fraction of its own 512 control positions that reaches the trigger – no cell count enters it and no form is assumed for the tail – and summed over the windows 36.5 are expected to trigger by chance. How many of those would be attributed is measured on the channels the control maxima land on. The channel of a maximum exists only for the 8 control positions per window that retain a dynamic spectrum (§4.5), so the share is measured on the 3 192 such positions of 399 of the 405 windows, which carry 98.5 per cent of the expectation above. The full ensemble holds 18 669 exceedances, which divided by 512 is the 36.5 above; the 8 are indistinguishable from a uniform draw from their window's own 512 by rank ($P = 0.56$) and their exceedance rate from the full ensemble's by +1.0 standard deviations. 88 of 302 exceedances fall inside the mask, carried to the star's frame exactly as a stellar cell, so the attribution takes 29 per cent of the expectation where a uniform landing would take 20. The atmospheric screen (Appendix E) then takes 10 of 214, leaving 24.8 expected unattributed Class A crossings against 31 observed, a

sum over rows with nothing carried at an assumed rate. Both sides are read over the whole primary beam §3.1 defines, with each block on its own antenna geometry; the narrower field of §4.3.1 disagrees with it on 6 of the 405 windows and in both directions: the whole beam holds a crossing the narrower field does not toward HR 1010 and HD 139084 B, both attributed to carbon monoxide, and the narrower field one the whole beam does not toward τ Cet, HR 1010 (twice) and η Crv. System by system 2 of 61 fall below one chance in twenty against 3 expected; the 15 untabulated archival-tail windows, with the two unattributed crossings toward HD 14055, enter neither side.

A count is compared with the distribution of counts chance delivers. Over 200 000 resamplings – each window's rate and attributed share redrawn from its own probes, and the execution blocks resampled because windows share them – a count of 31 or more arises with probability 0.14, conditional on the control-position model and not on an independently established null. Added to each resampled rate, 17 further crossings still leave 31 credible at 0.055 and one more does not (0.041 against 5 per cent). The uncertainty in 24.8 is carried inside that bound by the same resampling.

The one assumption the searched data cannot test is that a star behaves like a control position in its own field. The star carries 1.104 times a control position's cells in the band $4 \leq T < 5$ (1.046–1.161 at 95 per cent, blocks resampled). Calibrating the stellar rate against that ratio instead, the sensitivity test of §4.3.1, lowers the bound to 14; the whole-beam conclusion that there is no significant excess is unchanged under it, and the control positions' own rate is adopted as the smaller of the two and the more demanding of the count under test. The 31 would stand above chance at 5 per cent only if stars triggered at 0.89 times a control position's rate or less, and every ratio measured is an excess in the census. The reserved blocks trigger in 6 of 62 windows against 6.3 predicted, 5 of them in blocks sharing a session with a searched one, and the 28 sharing none hold one trigger event against 1.7 expected, so they neither confirm nor contradict the ratio (Appendix B.1).

Near a band edge the exceedance rate is $\times 2.464$ the interior rate, pooled – and the excess is one object: the window along the sightline to HD 48370 contributes 59 per cent of every near-edge exceedance in the survey, without it the ratio is $\times 1.020$, and on a held-out half it is $\times 1.181$ ($p = 0.24$). No edge correction is therefore applied to any limit.

D.2. *Execution-block quality*

The comparison assumes each block's control ensemble behaves like noise, and one block makes that false: all 512 of its probes exceed the trigger, the smallest at 6.6. Every block is therefore judged on the median of its windows' control ensembles divided by its class median, which also flags 5 of the 1663 windows individually at twice the survey median. *The criterion is post hoc*, written after the condition was seen; it is trustworthy because the failing block sits at 4.1 and the next at 1.12, a factor 3.7 apart, and all 41 thresholds between select the same block. 1 of 407 census blocks fails; its 4 coarse-channel windows hold the entire Class B crossing population, 4 of the 56, listed in the ledger and entering no count elsewhere.

D.3. *What the ensemble can resolve, and in what unit*

Ranking the star against 512 control positions decides which windows deserve examination, but $1/(512+1) = 1.95 \times 10^{-3}$ is the ensemble's *resolution*, not the probability that noise produced the window: a survey of 1663 windows would need a rank 65 times finer, and correlated controls inside the beam leave an annulus resolving no better than 0.023–0.032. That floor is the design's central limitation: no window can be significant on the rank alone, which is why confirmation requires independent recurrence.

The widened recurrence window of §4.6 is counted in windows, a window covering several crossings being one measurement, so the 328 pairings are 212 distinct windows, each one's chance formed from its independent channel–drift cells (10.4, $P = 0.03$, with its control positions supplying the rate instead; the larger P is quoted). Of the 17 windows above the trigger, 10 are toward HD 14055 and ALMA J153702653-33192492 alone, and over the other 161 windows the comparison is 7 against 4.8 at $P = 0.19$; those two hold 51 of the windows and 6.5 of the expectation, and §4.7 splits the departure between them. The full treatment is in `documentation/appendix_D_false_alarm.md` of the data archive (White & Dey 2026).

D.4. *The registration of the predicted cell*

The recurrence test of §4.6 reads one channel at one drift, registered between epochs to 0.75 of a channel in every window, the frame correction moving it by up to 116 channels. Widening each reading to the worst cell within $0.45\,\mathrm{km\,s^{-1}}$, pinning it at every anchor along the discovery track, and raising the bar to 4.1 to hold one false confirmation over all 444 readings each changes no disposition. What the recoveries establish about the registration of a single cell rests on the 3 recovery readings of HD 285968 alone, whose line is confined to one channel and which register to 0.06 of a channel, $0.038\,\mathrm{km\,s^{-1}}$; the rest test the frame correction only to the width of a resolved line, up to 69 channels.

D.5. *From the widened window's exceedances to the pairs*

Two counts of §4.6 are counts of different objects. The 29 above-trigger readings are window readings, one per pairing of an unattributed crossing with a window covering its frequency in another block – 328 such pairings over the 37 crossings a later catalogued block covers, row 2.3.3 of Table A2 – so a pair of unattributed crossings is read twice: 11 readings are second readings of 18 pairs, 2 outside the interval the pair test admits and 16 inside it, the set of Table 5, the other 3 of the 16 wanting a discovery spectrum or a fitted drift. The accounting for each pair is in the data archive (White & Dey 2026), `candidates/drift_pair_accounting.csv`. A catalogued window covers 32 of the 33 crossings that carry an amplitude bound and spans the orbital excursion of §4.6 in 27; for 5, toward β Pic, HD 110411 and HD 69830, the widest covering window is narrower, and 1 – CD−57 1054 – is covered only by the uncatalogued tail block `Xd09670_X3077`, 38.8 d later. Of the 7 pairs whose epochs are close enough together for the predicted offset to be tighter than the searched window, every one

TABLE D1
The crossings whose fitted drift lies on an outermost trial of their window's own grid, with that grid. Rates are in the observatory frame.

Star	block	ν (GHz)	$\dot{\nu}_{\rm obs}$ (Hz s^{-1})	trial	$\lvert a_{\rm los}\rvert$ (m s^{-2})
BD+05 1668	`X28a8`	242.195	+3452	2/2	≥ 4.27
HD 14055	`X29ae`	331.156	−3976	0/135	≥ 3.60
HD 14055	`X2c3bb`	345.526	+4184	141/141	≥ 3.63
HD 161868	`X10ca8`	345.634	+4195	127/127	≥ 3.64
TRAPPIST−1	`X259b`	346.245	−4623	0/137	≥ 4.00

fails a drift changing at a constant rate, by 3.6 times that resolution in the mildest case. The control row is a line that does not drift: the CO(2–1) line of HD 285968 is recovered at one stellar-frame frequency 5.0 days apart, within 0.6 of the comparison's own resolution of the prediction, so between these two blocks it resolves a mean drift to 51 Hz s^{-1}: a grid step carried over the gap, which is why a longer gap does not sharpen it.

β Pictoris is the closest case, and it is the one the comparison cannot dismiss; its orbit is argued here and nowhere else, the methanol identification of its two crossings being excluded in Appendix C. Its two crossings, 106 km s^{-1} apart in the star's frame in blocks 24.4 h apart, have negative fitted drifts (−1611 and −679 Hz s^{-1}), so a single carrier would have had to rise in frequency between the epochs and under constant acceleration misses by 132 times that resolution – but the mean acceleration it needs, 1.21 m s^{-2}, is inside the 3.6 m s^{-2} ceiling. The pair fixes three numbers and a circular orbit about a 1.75 $M_\odot$ star (Crifo et al. 1997) has three parameters: one orbit (1.54 d, 16.5°, 0.0314 au) reproduces all three, eccentric ones fit to $e \approx 0.28$, and the pair is not separable.

D.6. *The drift grid's edges*

The drift trials run from $-\dot{\nu}_{\rm max}$ in half-channel steps to the first trial at or beyond $+\dot{\nu}_{\rm max}$, so the searched interval is asymmetric by up to one step; 5 of the 56 crossings with a fitted drift lie on an outermost trial (Table D1), toward BD+05 1668, HD 14055, HD 161868, TRAPPIST−1, and are reported as lower bounds, 3 of them carrying no amplitude bound, which is why the table reaches beyond §4.6's range.

E. RADIO-FREQUENCY INTERFERENCE SCREENING

Interference is the first alternative explanation for an unattributed crossing. An interferometer is a poor detector of it – a signal not in the far field at the phase centre is suppressed in the correlation by its rapidly varying geometric phase – though allocations exist across the searched bands (International Telecommunication Union 2024), and the documented hazard, spaceborne cloud radar at 94.0–94.1 GHz (Stephens et al. 2002; Wehr et al. 2023; Emerson 2004), overlaps 1 of the 1663 census windows, with 0 crossings and 0 rank-passing events. *The tests of Table E1 carry none of the signatures interference leaves*, but their populations differ, each needing a different quantity recorded, and since the correlator flagging was not retained *no per-channel interference flag exists, and none can be recovered.* The one concentration a reader will notice, 12 crossings between 344.2 and 346.4 GHz, is the carbon monoxide tuning, 9 of the 40 unattributed crossings the census and the reserved hold-out together place in a searched window (row 2.4 of Table A2) against 8.8 ± 0.8 expected from where the windows were tuned. The image-sideband test is a bound, not a clearance, and 0 crossings lie where the receivers are double-sideband. The full treatment is in `documentation/appendix_E_rfi.md` of the data archive (White & Dey 2026).

TABLE E1
The interference tests, each of which could have failed. p is against the same peaks redrawn on their own channel grids.

Test	Population	Result	p
Clustering in sky frequency	51 census crossings with a released peak	11 within ±50 km s^{-1} of a searched transition against 4.8 expected: they cluster on the survey's own lines	0.0010
Crowding on one another	34 unattributed, quality-passing, of the census	no crowding at any of six separations from 1 to 300 MHz, where the 13 attributed crossings crowd at every one	≥ 0.08
Atmospheric transitions	62 crossings of the census and the hold-out, against 570 telluric lines	1 inside the 62.5 MHz screening radius in the topocentric frame, flagged as atmospheric and removed from both sides of the chance comparison while remaining among the unattributed; 1 inside ±50 km s^{-1} against 2.0 expected	–
Fixed intermediate frequency	1401 census windows with a released peak frequency	peak positions within their own windows uniform, median 0.50	–
Cross-target occupancy	3027 extracted windows toward 133 stars, every block set of Table A1 together	569 cross-star coincidences within 5 MHz against 554 ± 13 expected; *below* the null over the 109 above-trigger windows, 4 against 5.2 ± 1.9	0.15
Image sideband	the 36 unattributed crossings lying in a catalogued block	28 have no catalogued transition in the image interval and 4 have one the same block observed with nothing bright enough to leak; for 4 the parent frequency was not observed	–

One execution block fails these tests, as a block and not as a carrier. Its 4 crossings toward BD+05 1668 are the 4 highest in the survey, none below $T_\star = 39.6$ (§4.3.2), and its 24 sibling executions produce no crossing. *None of this substitutes for the ON/OFF cadence a dedicated search uses*: these are statements about the population, not a clearance of any one crossing.

F. PER-SYSTEM SENSITIVITY

The limit Table F1 quotes for a system is that of its best window, a bound pair appearing under its primary's name. A window inside the attribution mask constrains nothing (§3.2) and cannot be a best window: a double dagger marks a row keeping less than half its channels, and HIP 10679, whose 2 windows are all inside it, is tabulated with dashes. Beside each limit are that window's centre frequency ν and channel width $\Delta\nu$, the system's merged coverage $\Delta\nu_{\mathrm{cov}}$, the percentage *dec.* by which atmospheric decorrelation leaves the limit optimistic, and whether blocks of different sessions cover a common frequency more than a day (d) or a year (yr) apart, over the census and the 166 blocks the recurrence test reads. *The decorrelation column is measured where the delivery allows it and imputed where it does not*: an asterisk marks the 32 systems whose best window lies in a block with no phase record of its own, imputed from the blocks matched to it in baseline, against 27 measured, with a dash for the one system whose blocks offer neither. The 3 per cent of §4.1 is therefore a mixture: 4 per cent over the measured systems and 3 over the imputed, whose estimate is a median over matched blocks and so has little spread of its own. Those blocks are not a random subset – 54 per cent 12 m against 26 per cent of the ones that do expose a record, that array's long baselines being what loses coherence – so each of their windows is matched to measured blocks within a factor of two in baseline length: 1.9 per cent in the median of those windows, and 21 per cent had each performed like the ninetieth-percentile block matched to it, which is a condition on a calibration performance the archive does not report. Every value in the column is a floor: the phase rms is recorded on the calibrator, not on the target. A dagger marks the 6 of the 60 systems whose own best window was one of the 49 injected into, so that their completeness is measured; the other 54 limits are transferred from the class average and carry instead the dispersion observed between injected windows, $\times 0.80$ to $\times 1.58$, and for the 8 systems whose best window lies outside the injected range in band, channel width, array configuration or integration time that dispersion is itself transferred. Drawing each window's factor from the 49 measured ones instead, over 4 000 trials, shifts the per-system median by 5.5 per cent of the adopted 3.03, which is how far a minimum over estimates favours low factors. The full treatment is in `documentation/appendix_F_per_system.md` of the data archive (White & Dey 2026).

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TABLE F1

Per-system sensitivity of the carrier search. For each of the 61 systems the fine-channel search covers, the $EIRP_{90}$ of its best window with its own primary-beam, annual-parallax and smearing retentions applied, ordered by the limit. Columns, daggers and dashes are defined above.

System	$EIRP_{90}$ (10^{14} W)	ν (GHz)	$\Delta\nu$ (kHz)	$\Delta\nu_{cov}$ (GHz)	dec. (%)	conf.	System	$EIRP_{90}$ (10^{14} W)	ν (GHz)	$\Delta\nu$ (kHz)	$\Delta\nu_{cov}$ (GHz)	dec. (%)	conf.
LP 876-10	0.87	345.83	488	1.7	4*	d	HR 1010	19	230.53	977	2.8	1	d
AU Mic	1.2	230.56	488	3.5	3*	yr	LHS 1140	19	230.00	488	1.8	0.4*	d
τ Cet	1.2	230.55	488	1.8	0.4*	d	κ CrB	19†	230.57	977	1.7	2*	–
HD 69830	1.2	230.06	488	3.8	1	d	HD 172555	21	340.94	488	5.3	200	–
61 Vir	1.3	345.79	488	1.8	5*	yr	HD 155555C	21†	346.01	488	1.7	43	–
ϵ Eri	1.4	229.98	977	2.3	–	yr	L 836-122	22	346.00	488	1.7	1	–
TRAPPIST-1	1.6	346.58	488	1.7	5	d	CD−57 1054	23	345.99	488	1.7	20	d
β Pic	1.7‡	226.85	122	4.4	2*	yr	CP−72 2713	24	345.12	488	5.8	64	d
AT Mic AB	1.7	346.04	488	1.7	3	–	HD 14055	24	230.11	488	7.2	3	yr
GJ 3379	2.2	229.96	488	1.9	0.4*	d	TWA 7	24	345.99	244	0.87	26*	d
GJ 14	3.1	227.22	488	3.5	19	d	GJ 2006A	24	346.02	488	1.7	1	–
HN Lib	3.9†	229.99	488	1.8	0.4*	d	Wolf 219	26	230.13	488	1.7	2*	d
HD 10647	4.4	345.79	488	6.1	0.8	yr	LP 736-15	28	345.82	977	1.7	4*	–
η Crv	4.5	343.35	488	7.2	27	yr	TYC 2211-1309-1	28	346.00	488	1.7	5	–
HD 202628	5.4	230.51	488	1.8	4*	yr	UCAC3 176-23654	29	346.02	488	1.7	4	–
HD 53143	5.4	230.53	977	1.8	2*	yr	PM J03433+1958	31	230.13	488	1.7	2*	–
HD 207129	6.4	230.57	488	1.8	3*	yr	UCAC2 20312880	32	345.96	488	1.7	11	–
GJ 849	6.8	230.02	488	1.9	0.4*	d	LP 353-51	33	346.02	488	1.7	26	–
TWA 3A	7.6†	230.54	122	0.44	35*	d	Barta 161 12	34	346.02	488	1.7	4*	–
HD 139664	7.8	230.55	488	1.7	3*	–	51 Eri	37	230.53	1953	1.7	11*	–
HD 92945	8.3	355.11	488	3.7	6*	d	HD 23484	37	230.53	977	1.8	0.2	d
TX PsA	8.5	346.03	488	1.7	7	–	HD 139084	54	346.01	488	1.7	3	–
HD 285968	9.1	229.97	488	1.8	0.4*	d	WD 0407+179	84	230.13	488	1.7	2*	–
HD 161868	9.9	227.22	488	7.4	5	d	HD 48370	88†	230.73	122	3.3	0.4*	yr
HD 38858	13	230.50	488	1.8	3*	yr	HD 45184	93	230.53	977	1.7	2	d
HD 110411	13	219.58	244	4.7	2*	yr	ALMA J153702653-33192492	99	345.98	488	14	5*	d
HIP 490	13†	230.53	244	0.86	3*	–	HD 31392	130	230.52	977	1.7	0.9	d
HD 107146	14	355.14	488	3.7	6*	d	HD 89452	260	230.53	977	1.7	0.7	d
LP 476-207	15	346.02	488	1.7	4	–	HD 377	370	230.54	977	1.7	68	d
HD 170773	17	230.13	488	7.4	2*	d	HIP 10679	–	–	–	0.11	–	–
HD 61005	19	230.52	244	19	3*	d							

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